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Received: 7 December 2021 | Revised: 28 January 2022 | Accepted: 28 January 2022

DOI: 10.1002/pat.5636

REVIEW ARTICLE

polymers advanced technologies WILEY

Oil spills adsorption and cleanup by polymeric materials: A review

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Funding information
Ministry of Science and Technology, Taiwan, Grant/Award Number: MOST-108-2221-E-035-054-MY3

Abstract

Oil spills are environmental disasters that impact human, plants, and wild life, such as birds, fish, and mammals. This article has comprehensively reviewed the progress made in oil spills remediation especially by polymeric adsorbents. Various methodologies involved in oil sorption using porous polymer network and its recovery are thoroughly discussed. In addition, new concepts and development in synthesizing sponge, fiber, graft, and crosslinked polymer networks are also collected briefly and future perspectives are suggested. With their unique physical and mechanical properties coupled with their very high surface area and small pore sizes, polymer-based adsorbents can be used as advanced materials for cleaning up oil spills.

KEYWORDS
nanopolymers, natural sorbents, oil spill cleanup, porous materials, super adsorbent polymers

1 | INTRODUCTION

Ocean provides a vibrant contribution in maintaining natural resource,[1] transportation,[2] climate and weather,[3] and economy.[4] Most of the earth's natural oxygen is a product of photosynthesis in ocean organisms, which absorb carbon from the atmosphere.[5] Majority of the world's population lives within the coastal zone and the transactions by mainly ocean-related businesses contribute more than $500 billion to the world's economy. Directly or indirectly ocean provides a million of jobs in industries, such as fishing,[6] aquaculture,[7] and tourism.[8] In addition, the ocean provides a great deal of contribution in the medicinal field, where the utility of sea creatures in the form of ingredients helps to synthesize novel medicinal compounds to treat cancer-related issues.[9] The majority of people simply depends on the ocean in order to fulfill their needs for their survival, but unfortunately sometimes they do unwise practices that have created serious issues, such as over reaping,[10] loss of biodiversity,[11] and deprivation of marine habitats and creatures. In the meantime, transportation and dumping activities through shipping could create pollution and high complexity to the marine ecosystem. These have impacted on reproduction, feeding, and respiration, further led to oxygen deficiency and killed ocean organisms.[12] Plastics and garbage thrown into the sea can harm the marine life and sometimes it gets trapped on propellers in seawater intakes.[13]

Recently, environmental issues arising from oil pollutions are becoming complicated because it not only worsens the human health but also deteriorates the ecological environment.[14–16] The present fast development of the modern economy with a high demand for energy has provoked oil exploitation and transportation, which results in oil spills.[17] World's largest percentage of energy consumption are satisfied by oil, ranging from a minimum of 32% for the Europe and the Asia up to a higher 53%, and for each the America and the Africa, which consumes approximately 40%.[18] Thirty billion barrels of oil per year are consumed all over the world and most of the countries are simply depending on developed nations for their oil demands.[19] According to historical statistics, 24% of the oil consumed in 2004 went to the United States alone.[20] Some of the developed nations, who are rich in their oil productions are responsible for 60% of barrels flowing each day.[21] Saudi Arabia is the first country to produce 11.75 barrels of oil per day and crowns as oil champion.[22] Following them, the United States (10.59), Russia (10.3), China (4.19), Iran (4.13), Canada (3.92), United Arab Emirates (3.23), Mexico (2.95), Brazil (2.8),

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and Kuwait (2.75) are among the top 10 oil-producing countries in the world.[23] These top 10 producers are responsible for over 60% of the barrels flowing each day. Almost 90% oil-related releases occur during transport from one region to another via ocean. Marine oil spills are generally much more damaging than those on land, since it can spread for hundreds of marine miles in a thin oil slick, which can cover beaches with a thin coating of oil.[24] The oil contaminant contains molecules that have low density and are more hydrophobic in nature, hence cannot dissolve in water and form a thick sludge.[25] Such accidents can worsen the marine life by creating serious damage to the coastal environment and long-term harmful impacts on aquatic ecosystems. The cleanup of this contaminant from the ocean is a major challenge that entices a great deal of attention. Huge number of oil spill accidents occurs in the past five decades.[26,27] Some of the historically important oil spill incidents that are reported in the literature are summarized in Tables 1 and 2. Deepwater Horizon Oil spill accident occurs in 2010, reported that nearly 4.9 million barrels of crude oil were released in the Gulf of Mexico, which is considered to be the largest marine oil spill in US history.[28] This terrific incident subsequently created a huge damage to habitats around the ocean. Exxan Valdez oil spill occurred in 1989, which spilled over 11 million gallons of crude oil into the marine.[29] It came next to deep water horizon oil spill and was considered as a largest single oil spill in US coastal water. Ixtoc I oil spill occurred in 1979, reporting 3 million barrels of oil spilled in the Gulf of Mexico to cover a 2800 km^2 area and a shoreline impact of nearly 261 km.[30] A majority of oil spill accidents mainly falls in the early 80s, namely Atlantic empress (1979) (West Indies), Nowruz oil field (1983) (Persian gulf), Castillo de Bellver (1983) (South Africa), Amoco Cadiz (1978) (France), and Odyssey (1988) (Canada).[31] All these incidents released approximately 80 million gallons of crude oil causing a huge economic loss and adverse effects on coastal and other ecosystems. According to energy-related severe accident database out of the total 1213 accidental oil spills considered all over the world, the two largest spills, namely Deepwater Horizon and Ixtoc I (1979, 480,000 tons), account for nearly 11% of the total oil spilled and the 50 largest spills account for 61% of the total amount.[26] Even though the oil spill incidents continued over the decade, the amount of oil spilled and the number of spills greatly get reduced according to statistical reports. Proper cleaning measures are necessary to separate oil from water to protect our environment and ecosystem.[32]

2 | PRINCIPLES OF OIL IN WATER SEPARATION

Oil/water separation is a field of high implication as it has direct practical consequences for resolving the problem of industrial oily wastewater and other oil/water pollution. Oil in water is a heterogeneous mixture and can be separated into their components by several physical/chemical methods.[33] When a mixture of oil and water is left undisturbed for a period, a continuous layer of oil forms at the surface of the aqueous phase if the oil is less than water. Separation of the oil/water mixture into two separate layers is due to the action of flexible forces on the large oil masses initially present in the water frame. The oil in the upper layer has external physical and chemical properties of the source oil, unless it has been modified by the reaction with chemicals present in the aqueous environment.

In oily water mixtures, the oil exists as a collection of beads of various sizes. The force acting on such a globule, which causes it to move in the water, is proportional to the difference in weight between the oil particle and a particle of water of equal volume. Oil droplet in oil/water emulsion (Figure 1) simply depends on the phase volume ratios and other factors. The particle of the dispersed oil in an emulsion is important in characterizing the dispersion, which usually has a very small fraction of oil in the order of 50–1000 ppm.[34]

Adsorption is a surface process in which the adsorbed molecule does not diffuse into the lattice of the solid. Adsorption by adsorbent materials have attracted huge attention, such that selection of a proper adsorbent is very important for these processes.[35] Several studies have been carried out on vital aspects (e.g., adsorption capacity and mechanism) of the separation of oily species from water streams. It has been proved that the adsorbents have a significant capacity for oil uptake from the water surface to reduce the financial plan and minimize harmful effects on ecosystems. In general, good adsorbent materials have some properties, including high hydrophobicity and oleophilicity, and efficient uptake capacity.[36] The floating oil can be transformed from a liquid to a semi-solid, and then be removed from the spilled area by an appropriate mechanical technique.[37]

The adsorption capacity of an adsorbent is a function of the surface area of the pores in its body. Generally, the internal surface area of pores is several orders of magnitude larger than the outer surface area of the adsorbents. Almost all sorbent materials have the capacity to adsorb and hold water vapor, but selective sorbents, such as silica,[38] alumina,[39] zeolites,[40] and so forth, have significant capacity for holding the oil by suitable chemical modifications. These sorbents adsorb large amounts of molecules into the pores of their surface. This is made by the fact that these pores have a large active surface area in the range of 500 m^2/g due to its unique internal structure and a vast network of interconnected microscopic pores.[41] The important properties, such as surface area, pore volume, pore size distribution, and contact angle, are among of which they are strongly associated to the adsorption capacity.[42] Large surface area and high pore volume

TABLE 1 Number of oil spills happened through different sources during 1970–2020

	Storage/refinery	Exploration and production	Ship	Pipelines	Total
No. of oil spills accidents	113	24	888	188	1213
Total of oil spilled (tonnes)	870,000	2,200,000	6,000,000	750,000	9,800,000

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TABLE 2 Quantities of oil spilt 7 tonnes and over (rounded to nearest thousand) 1970–2020

No.	Category name	Location	Year	Spill type	Amount in tonnes
1.	Deep water horizon	Gulf of Mexico	2010	Crude oil	470,779
2.	Atlantic empress	Off Tobago West Indies	1979	Crude oil	287,000
3.	Castillo de Bellver	Off Saldanha Bay South Africa	1983	Light crude oil	252,000
4.	ABT Summer	Off Angola	1991	Heavy crude oil	260,000
5.	Ixtoc I well	Gulf of Mexico	1979	Oil spill	470,000
6.	Sea empress	Milford haven UK	1996	Crude oil	72,000
7.	Exxon Valdez	Prince of wales island, Alaska	1989	Crude oil	37,000
8.	Khark 5	Canary islands	1992	Crude oil	80,000
9.	Erika	Bay of Biscay France	1999	Heavy fuel oil	20,000
10.	Prestige	Off Spanish coast	2002	Heavy fuel oil	64,000
11.	Hebei spirit	Yellow sea south Korea	2007	Crude oil	10,800
12.	Pipeline	Ahrazin Iran	1978	Oil spill	95,240
13.	Amoco Cadiz	Brittany France	1978	Oil spill	223,000
14.	Odyssey	Nova scotia Canada	1988	Crude oil	132,000
15.	Torray conyan	Isles of scilly England	1967	Crude oil	119,000
16.	Sea star	Gulf of Oman	1972	Crude oil	115,000

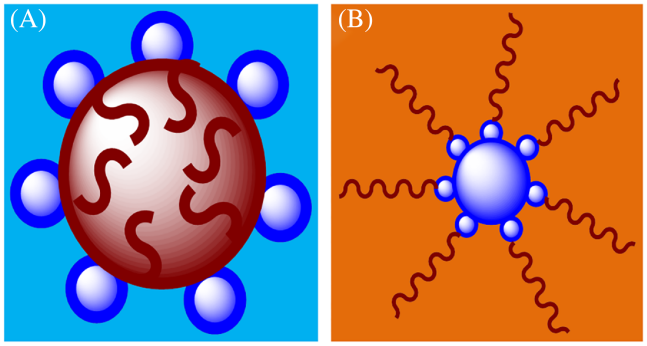

FIGURE 1 Graphical illustration for the formation of emulsions (A) oil in water (B) water in oil

are widely used in chemical and oil separation, where the total surface area may support the accessibility of active site. Pore size distribution with the combinations of the micropores (pore diameter < 2 nm) and mesopores (pore diameter 2–50 nm) are required to improve the transport process of particles or molecules inside the porous networks and facilitate the better adsorption. In addition, the contact angle[43] (the angle formed by the intersection of the oil–water interface) gives a better idea about the wettability of the sorption surface. Small contact angles (80–90°) correspond to high wettability, when the liquid spreads on the surface. Similarly, large contact angles (140–150°) correspond to low wettability, when the liquid beads on the surface. More specifically, a contact angle less than 90° indicates that wetting of the surface is favorable and the liquid will spread over a large area on the surface. While contact angles are greater than 90°, it generally means that wetting of the surface is unfavorable, so the liquid will minimize its contact with the surface and form a compact liquid droplet. However, complete wetting occurs when the contact angle is 0° as the droplet turns into a flat plate. For super hydrophobic surfaces, water contact angles are usually greater than 150°, showing almost no contact between the liquid drop and the surface, which can rationalize the water in lotus leaf.[44]

3 | SORBENTS FOR DEEP MARINE OIL SPILLS

Molecules of nonpolar compounds, such as oil and organic solvents, even when mixed well into water, tend to separate from the water, while the mixing stops. Water molecules tend to hold on to each other and squeeze out nonpolar oil and gasoline. Because of density differences between water and oil, they form two separate liquid layers. The lower density nonpolar components can easily separate out from water, which can capture effectively by sorbents in comparison to high density polar solvents. High density polar solvents that can effectively miscible with water molecules and the amount of liquid adsorbed can be effectively reduced. In contrast, the mechanical extraction by the use of suitable sorbents can be considered as one of the most effective method for emulsion removal due to its ease in handling and implementation in contaminated areas. The adsorbing phase is the adsorbent and the material concentrated or adsorbed at the surface of that phase is the adsorbate. The adsorbent materials are necessarily high-performance smart materials having super hydrophobicity and superoleophilicity,[45,46] which removes oil without leaving a trace of secondary pollution. Additionally, mechanical recovery using sorbent material is highly appropriate due to the absence of risk concerning, an additional environmental contamination. The ideal sorbent material for an oil spill should satisfy the following requirement that includes super hydrophobicity, high uptake capacity and rate,

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resistance, retaining ability, durability in aqueous media, ecofriendly, biodegradability, and recoverability of the absorbed oil.[47–49] Various adsorbent materials, which are used as an effective oil spill separators have been reported in the literatures that include zeolites,[50] carbon material,[12] keratin,[51] clay,[52] sugar derivative,[53] nanoparticles,[54] and others. These novel materials had a great impact in separating oil and water in different ways.

4 | NANOPOROUS MATERIALS AS ADSORBENTS

Nanoporous materials are vastly beneficial materials and have wider applications in various industrial outcomes due to their unique physical and chemical characteristics.[55] It consists of either an organic and/or inorganic framework, which maintains a porous structure with a typically large surface area in excess of 400 m^2/g.[56] It also had a wide variety of applications ranging from catalytic reactions, adsorption, and environmental processes due to its architecture at the nanometer scale.[57] Owing to its increased surface area, higher reactivity, and the possibility of in-situ treatment, these nanoporous materials are more effective than conventional techniques, which offer a significant promise for oil spill cleanup and recovery. Carbon materials had a greater impact for oil spills in recent years due to their outstanding performance factors, such as water uptake capacity, regeneration factor, and recyclability.[58] Hydrophobic core shell magnetic Fe_2O_3@carbon nanoparticle selectively absorbs oil from water—oil mixture, because it imparts both superhydrophobic and superoleophilic properties on the core shell structure.

TiO_2 is a highly efficient nanomaterial having a significant potential for photocatalytic degradation of crude oil fraction. TiO_2 nanoparticle has a limited application in the oil spill remediation due to its tendency to form agglomerates, resulting in the reduction of its photocatalytic activity. However, TiO_2 in combination with carbon materials expanded its applications due to its dual character, such as adsorption/absorption and decomposition, which plays a synergistic role in treating oil related problems.[59] The nanostructured materials made up of graphene/TiO_2 enhance the efficiency of solvent removal due to its easy transfer of electrons and further recombination of electron hole is significantly reduced. MnO_2-silicon membrane was prepared by hydrothermal method comprising super hydrophobic/oleophilic properties in it.[60] The membrane is more selective for oil sorption due to its super wetting mesh structure. The membrane is thermally stable and can be recovered easily by simple ultrasonication. Silica aerogel in combination with multiwalled carbon nanotube (MWCNT) shows excellent adsorption capacity for the removal of oil from oil/water mixture. The composite aerogel possesses compressive stress and hydrophobic in nature result in an increased surface area, which is more suitable for adsorption process.[61]

Natural clays containing mostly metallic cations, such as aluminum, silicon, and magnesium, are shown to impart hydrophilic in nature. However, suitable modification of these materials with amine derivatives resulting in organophilic clays render hydrophobic character and more selective in adsorbing the organic contaminants from oil/water mixture. Recently, a nanohybrid was successfully produced to treat oil spills effectively.[62] The nanohybrids comprise of iron oxide nanoparticle in combination with natural clay by in situ co-precipitation. This nanohybrid adsorbs nearly four times its weight from oil/water mixture. Furthermore, the recovery has been accomplished easily, since the material is magnetic in nature and it can be removed under the application of magnetic field.

Wang et al.[63] constructed a low drag porous ship with super hydrophobic and super oleophilic surface for oil spill cleanup. For that they used zinc oxide nanorod arrays on the surface of the porous stainless steel wire mesh together called porous unmanned ship (PUS) by hydrothermal method. About 94% of high oil recovery capacity, 31% of drag reducing property at a flowing velocity of 0.38 m/s, good corrosion resistance, and reusability, were reported for PUS due to their superhydrophobic and low water adhesion force. The mesh was coated with a layer of crosslinked polydimethylsiloxane (PDMS) to lower the surface energy to achieve the super hydrophobic and oleophilic property. Hu et al.[64] reported the MWCNT-graphene hybrid aerogels by microwave irradiation for oil spills. The MWCNTs are immobilized on the surface of cell walls of graphene aerogels, which in turn attains physical properties like super hydrophobicity, super oleophilicity, large pore volume, pore size, and excellent compressibility for recyclable oil sorption. The synergistic effect between graphene and carbon nanotube (CNT) makes it a convenient tool for environment remediation especially for oil sorption, which is superior to those with most super hydrophobic and oleophilic structure. Similarly, Bi et al.[65] recently reported a shape moldable spongy graphene (SG), which is a highly efficient nanoporous material with a high specific surface area and a sorption capacity. SG has a high degree of regeneration and full release of adsorbate, which is an efficient absorber for not only petroleum products and fats but also toxic solvents that holds 86 times of its own weight. Cong et al.[66] fabricate a macroscopic graphene iron oxide hydrogel for oil spill cleanup by the self-assembly of graphene oxide (GO) sheets with simultaneous coating of metal oxide nanoparticle. In the same manner, Wu et al.[67] also utilized GO sheets to produce graphene-based monoliths. The reduced GO sheets were assembled on the skeleton of polyurethane (PU) foams, which have a best remedy for oil spills. Hayase et al.[68] demonstrated a hydrophobic porous methyltrimethoxysilane-dimethyldimethoxysilicone gel by means of sol gel process. The elastic nature and thermal stability makes the reported gel more prone to oil/water separation compared to PU. Gao et al.[69] demonstrated a novel porous nitrocellulose membrane to ensure under water superoleophilicity with an oil contact angle of 156°. The material possesses high water permeation flux density and selective for oil/water mixture with high efficiency.

5 | POLYMERIC NANOMATERIALS AS SORBENTS

Chung and Yuan[70] investigated that all the existing methods for surviving with oil spills like the 2010 Deepwater Horizon disaster are

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outdated technology and have many disadvantages. For example, corncobs, straw, and other natural adsorbents can hold approximately five times their own weight and pick up water, as well as oil.[71] Disposing those materials is another headache, which become industrial waste that must be disposed in special landfills or burned. Even though most of the natural sorbents are more economical than most synthetic sorbents, it also suffers some drawbacks, including relatively low oil sorption capacity as mentioned before. So far, this demerit has been improved by the surface and pore-structure modifications. Nanomaterial surface modified with different functional features, such as chemical (hydrophobicity, hydrophilicity, and oleophilicity), magnetic, mechanical, and photocatalytic properties due to their synergistic effect, has enhanced capability for more efficient removal of oil spills from water surface.[72] However, these approaches somewhat limit their practicality. Over the past few years, polymers have received considerable attention based on their outstanding properties, such as high porosity, elasticity, and open celled structure. In addition, the properties of polymer can be fine-tuned by modification with nanostructured materials, which render superhydrophobic and superoleophilic characters that are essential for oil selectivity and oil adsorbent capacity. A polymer has a promising tendency that soaks higher percentage of oil forming a rubbery substance, which prevents oil slicks at sea.[73] The hydrophobic properties of the polymer cause it to float on water; however, the density of the polymer is great enough to allow it to sink through the hydrocarbon and enriches the binding. The polymeric material transforms an oil spill into a soft gel like substance, which is a combination of solid and oil. The material can recover gallons of crude oil and its gel-like nature helps to collect, transport, and recycle. Therefore, the polymer-based materials vividly diminish the environmental pollution from oil spills and help to recover our natural resources.

6 | ORGANIC POLYMER NANOCOMPOSITE SORBENTS

Synthetic polymers have been established to be active for the adsorption and removal of isolated and dissolved organic contamination from water. Magnetic-organic polymer nanocomposites have special response due to their external magnetic field in the separation and nontoxic behavior. An environmentally friendly oil separation method using polyvinylpyrrolidone (PVP) coated magnetic nanoparticles (NP) to separate oil from oil–water mixture was reported recently by Mirshahghassemi and Lead.[74] They developed a cheap and facile hydrothermal method to produce the magnetic nanopolymer at lower temperature without the use of any organic solvents (Figure 2A).

The effect of oil extraction capacity of NP was monitored through gas chromatography—mass spectroscopy measurements. Chromatogram peaks for all oils containing C9–C26, that is, from nonane to hexacosane samples, decreased significantly compared to the initial oil–water mixture showing excellent and rapid alkanes removal of NP (Figure 2B). Results indicate approximately 100% of lower chain alkanes (C9–C21) were removed by magnetic separation. For longer chain alkanes, magnetic separation greater than 67% of C22–C25 was removed. These results show high oil removal efficiency of NPs from oil–water mixture in the ultrapure water. From these results, we find that PVP–magnetite composite has the sorption capacity of 30 g/g and this method can be potentially used for immediate separation of oil from water. Authors claim that these nanoparticles can be utilized to remove oil over a short time with high removal efficiency under environmentally relevant conditions.

Fernandes et al.[75] reported a simple method to obtain oil sorbent magnetic micro capsules for oil capture and recovery by utilizing magnetic nanoparticle and poly ionic liquid (Figure 3). They encapsulate an aqueous solution of magnetic nanoparticles in conventional liquid

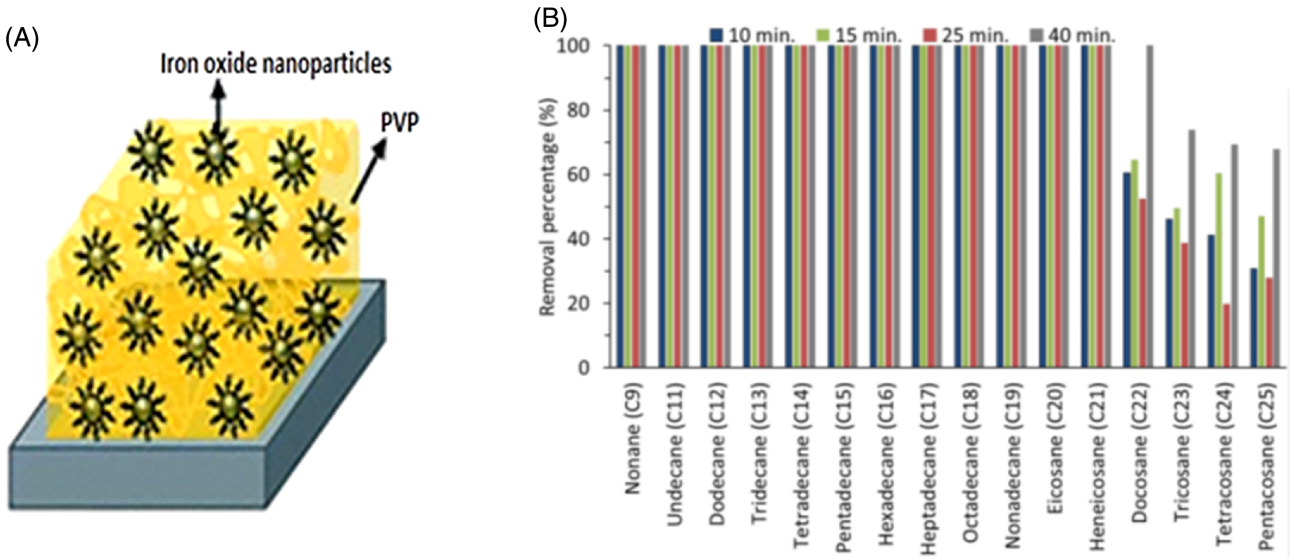

FIGURE 2 (A) Schematic representation of iron oxide/poly(vinyl pyrolidone) (PVP) oil sorbent. (B) Percentage of the solution remaining after oil removal using iron oxide/PVP sorbent. Reprinted with copyright permission from American Chemical Society[74]

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marbles based on poly(3-*n*-tetradecyl-1-vinylimidazolium bromide) nanoparticles. These microcapsules provide excellent hydrophobic and oleophilic surface properties and they were easily synthesized in a large scale via dispersion polymerization. The material shows almost 100% oil sorption behavior in oil/water mixtures, retaining their magnetic properties. The higher adsorption capacity value of 11 g of oil per gram of capsule was reported, which shows that most of the oil absorbed filled the capsule inside void and only a small part of it swelled in the capsule shell as reported by the authors. The magnetic property of the microcapsules promotes an easy recovery of oil after absorption. Hu et al.[76] reported highly efficient superhydrophobic magnetic foam based on polyethylene and magnetic titanium dioxide for oil spill treatment (Figure 4). TiO_2 was introduced into the magnetic foam to modify the surface morphology and chemical composition, exhibiting its excellent unsinkable property and improving the absorption capacity targeted at the oil/organic solvent. The uptake capacity of superhydrophobic magnetic titanium dioxide foam could reach about 64.31 kg/kg, providing excellent absorption capacity greater than 80 adsorption–desorption cycles, which shows its durability for almost all oils under study. Recovery can also be easily obtained by mechanical squeezing, which in turn avoids wastage of resources and proves to be an excellent remedy for oil spills.

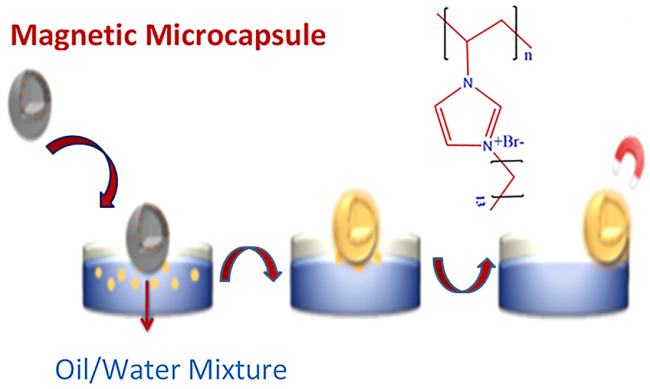

FIGURE 3 Chemical structure of poly(3-*n*-tetradecyl-1-vinylimidazolium bromide) and schematic illustration shows the method to determine the selectivity of a magnetic capsule to remove oil. Reprinted with copyright permission from Wiley online library[75]

Recently, Wooley and coworkers reported a robust magnetic polymer/hybrid nanoparticle for effective oil spill recovery by atom transfer radical polymerization (Figure 5A).[77] Well-defined magnetic shell-crosslinked knedel-like nanoparticles (MSCKs) were constructed through the co-assembly of amphiphilic block copolymers (BCPs) of polyacrylic acid-*b*-polystyrene and oleic acid-stabilized magnetic iron oxide nanoparticles. The hybrid micelles were prepared by combining oleic acid modified iron oxide nanoparticle and amphiphilic BCP and subsequent crosslinking with an agent resulted in the formation of MSCK. These hybrid nanomaterials were designed for application as sequestering agents for hydrocarbons. The presence of amphiphilic organic domains and capture of hydrophobic guest molecules renders MSCK system to achieve selective recovery of a broad range of environmental pollutants, including submerged oil and contaminated soils (Figure 5B,C).

Jiang et al.[78] reported a magnetic nanofibrous composite mat derived from polystyrene (PS)/polyvinylidene fluoride (PVDF) nanofibers with selective incorporation of iron oxide nanoparticle on PS via electrospinning process to treat oil spills. The oleophilic and hydrophobic properties were exhibited by the polymeric nanofibers and the presence of iron oxide (Fe_3O_4) nanoparticles have proved a

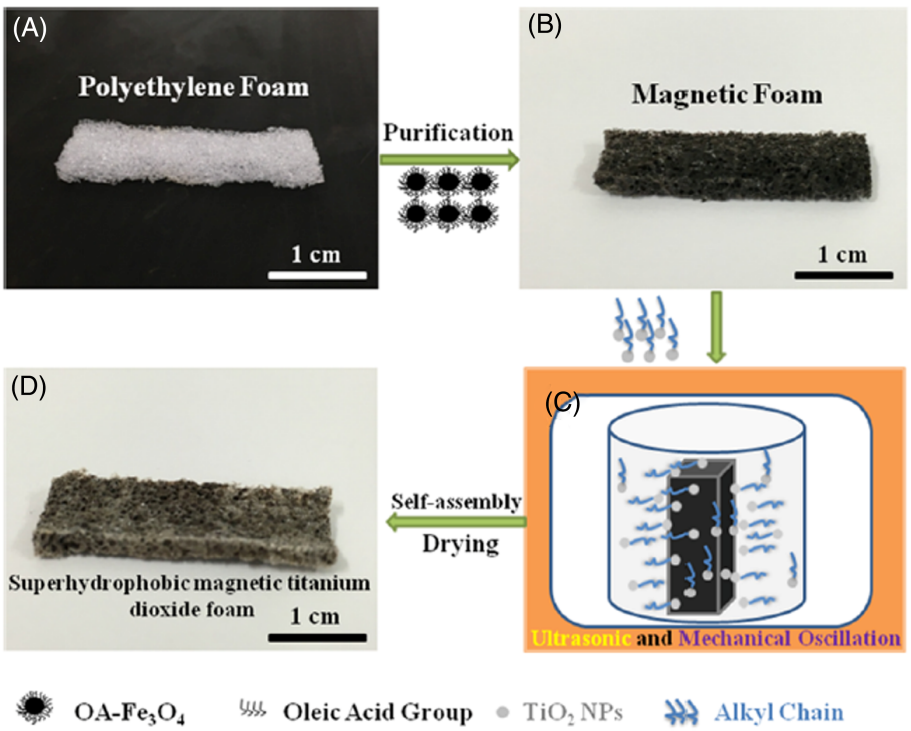

FIGURE 4 Photographic image and schematic illustration of the fabrication process of the superhydrophobic magnetic titanium dioxide foam. Reprinted with copyright permission from American Chemical Society[76]

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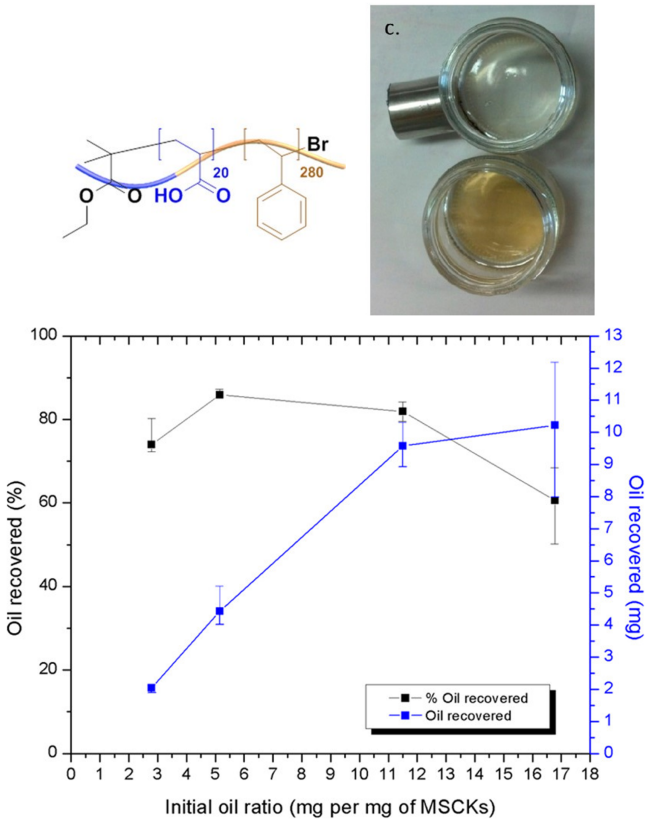

FIGURE 5 (A) Chemical structure of PAA_{20}-b-PS_{280} amphiphilic blocks copolymer and graphical representation of magnetic shell crosslinked nanoparticle. (B) Top view comparison of crude oil-loaded magnetic shell-crosslinked knedel-like nanoparticles (MSCKs) captured against the vial wall by an external magnet. (C) Oil recovered data in percentage and mass using MSCK. Reprinted with copyright permission from American Chemical Society[77]

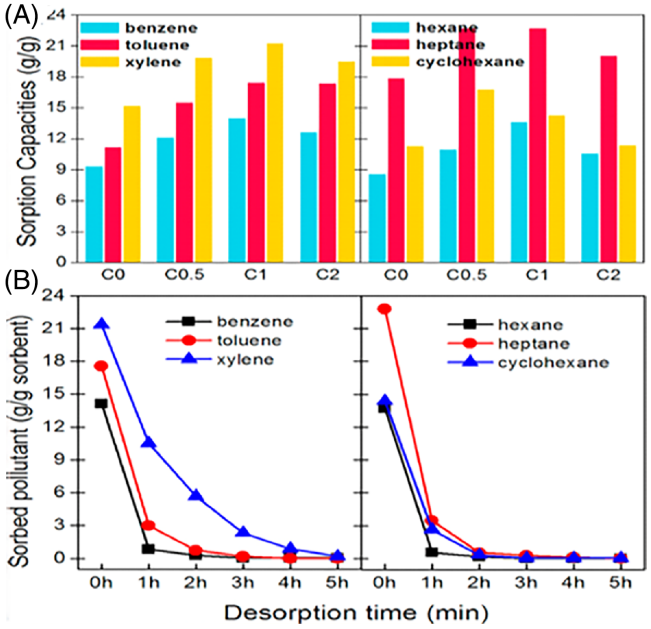

FIGURE 6 Sorption capacities and desorption curves of the graphite/isobutylene-isoprene cryogel for toxic organic liquids. Reprinted with copyright permission from American Chemical Society[79]

good magnetic property and helped on easy removal or separation of oil from the water surface. The oil sorption capacity of the materials was tested utilizing sunflower oil, soybean oil, motor oil, and diesel oil. The oil sorption capacity of PVDF/PS and PVDF/Fe_3O_4@PS sorbents were 44–46 and 35–36 g/g, respectively. The PVDF/Fe_3O_4@PS has a slight lower sorption ability, which may be due to Fe_3O_4 nanoparticles possibly blocking and reducing the porosity of the mat, however it provided a good solution for environmental remediation specifically on oil spills. Liu and coworkers[79] reported a new highly porous cryogel graphite/isobutylene-isoprene rubber by a cryogelation method as new sorbents for oil spills and organic liquids. Cryogels were sponge-like soft materials with excellent buoyancy and hydrophobicity. The cryogels were efficient sorbents for the removal of small oil drops dispersed in an emulsion. One gram of cryogel took up 17.8 g of crude oil, 21.6 g of diesel oil, or 23.4 g of lubricating oil (Figure 6). The samples also showed excellent sorption capability for aromatic compounds and alkanes, including 17.6 g of toluene, 21.4 g of xylene, and 22.8 g of heptanes per gram of sorbent, and recovery by squeezing or centrifugation. 3D covalent assembly of graphene/polypyrrole foam through hydrolytic condensation polymerization and hydrothermal reduction was studied by Liu et al.[80] (Figure 7A). The covalent assembly was enriched by electrostatic force and the interaction of pyrrole with the negatively charged —COO on the edge of the GO sheet was responsible for very high sorption capacities of the foam. The sorption was very fast and was completed within 5 min. The recycling performance of the diesel oil was shown to be 95% even after 10 cycles, indicating the good recycling performance of the foam for oil adsorption (Figure 7B). Chen and coworkers[81] reported the homogeneous polymer–protein hybrid oil–absorbing gel by treating acrylate and keratin-bromide by Living Radical Polymerization (Figure 8). They adopted the grafting from strategy, where the butyl methacrylate polymer chain propagated from the protein macroinitiator to prepare protein polymer conjugate. The effect of gelation and oil absorbency of the synthesized protein polymer conjugate can be monitored by using cross linkers, which could cause earlier gelation. The oil-absorbing property of different solvents was tested. Chloroform shows the highest value with saturated oil absorption of 43.7 g/g and contrastively *n*-hexane was the lowest of 2.7 g/g. Aromatic crosslinker was more prone to form micro-gel and higher oil absorbency was achievable with longer oleophilic chain of acrylate. The polymer–protein was obtained with the maximum oil absorption of chloroform reaching 43.7 g/g and possessed good reusability. This protein polymer hybrid with a suitable crosslinker offered a convenient way to prepare environmentally co-operated oil-absorbing material. A smart fiber membrane with two switchable wetting states made up of pH responsive BCP on a stainless steel by an electrospinning method was recently reported (Figure 9C).[82] The copolymer comprising of both methyl methacrylate and vinyl pyridine links obtained through Cu(0) mediated reversible deactivation radical polymerization

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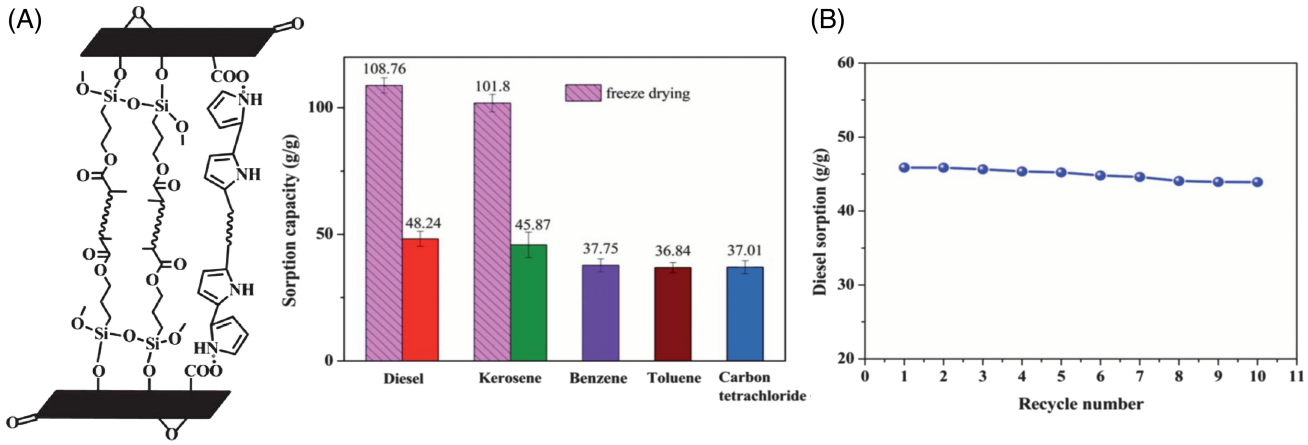

FIGURE 7 (A) Chemical representation of graphene/polypyrrole based foam obtained by free radical polymerization. (B) The sorption capacity and recycling performance of organic solvents using 3D graphene/polypyrrole. Reprinted with copyright permission from RSC[80]

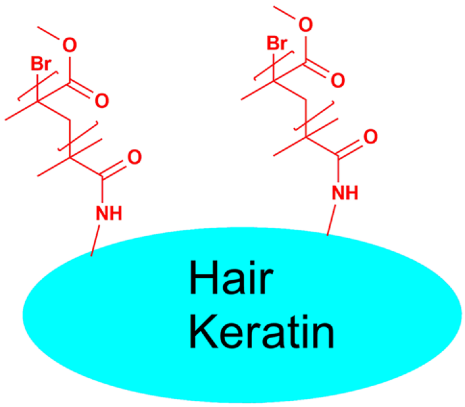

FIGURE 8 PMMA graft polymer from the surface of keratin by SET-LRP method. PMMA, poly(methyl methacrylate) Reprinted with copyright permission from Elsevier[81]

accomplished for better oil–water separation. Oil selectivity passes through the membrane and water remain at the initial state at a neutral pH, whereas after wetting the membrane with acidic water the reverse separation is emerged (Figure 9A,B).

Jiang et al.[83] in 2004 reported the modified stainless steel mesh with poly(tetrafluoroethylene) to incorporate superhydrophobic and superoleophilic surface, which could effectively separate the oil from oil/water mixture. Stainless steel mesh was also modified with poly(acrylic acid) hydrogel to check the efficiency in separating oil/water mixture. Apart from smart fiber membrane reported by Li et al.,[84] a similar work comprising thermo and pH responsive behavior was demonstrated by Cao et al. that they reported a stainless steel mesh modified with poly(2-dimethylaminoethyl methacrylate) hydrogel. The materials show superhydrophilic and under water superoleophilic properties at lower temp and pH, whereas it is permeable to oil at higher temperature and pH due to decreased swelling volume of the hydrogel coating. Xiong et al.[85] demonstrated unique responsive switchable fabric for water in oil emulsion. The zinc oxide nanorods were modified with poly(ethylene terephthalate) to display both underwater oleophobicity and under oil hydrophobicity, which was proved to be a simple in-situ tuning wettable filtration process for water/oil mixture. Ratcha et al.[86] reported a latex modified poly(alkyl acrylate) foam as an oil sorbent material prepared through latex vulcanization and cured by steaming. The latex obtained from natural rubber herea brasiliensis, which was used to prepare the foam for petroleum based liquid sorbents. Various poly(alkyl acrylate), namely poly(methyl methacrylate), poly(butyl methacrylate) (PBMA), and poly(butyl acrylate), were utilized for the preparation of the foam, which enhances the hydrophobicity/oleophilicity and decreases water sorption capacity with increased oil sorption. The oil sorption was tested with petroleum-based liquids, namely xylene, toluene, gasoline, diesel oil, and engine oil. The sorption capacities of PBMA modified natural rubber foam in the first cycle was approximately 16 g/g for xylene and toluene, 10.5 g/g for gasoline, 8.8 g/g for diesel, and 5.1 g/g for engine oil. Oil recovery efficiency of high viscosity liquids (diesel and engine) was 45%, which can be reused up to 30 times.

Essawy et al.[87] reported a novel interpenetrating amphiphilic co-networks based on compatibilized acrylonitrile butadiene rubber/styrene butadiene rubber (NBR/SBR) montmorillonite composites provide a possible solution to oil spills problems. The polymer network consists of polar acrylonitrile butadiene rubber and the non-polar styrene butadiene rubber and the compatibilization were accomplished by montmorillonite. The oil sorption and swelling process of the polymer network were tested with toluene and the capacity of oil uptake showed good correlations to the crosslinking density. The swelling percentage of the polymer network was found to be 180%, efficient for oil spills especially toluene. The ability of this kind of material can preserve low viscosity oils without rebleeding under small external pressure. A novel octadecyl acrylate (OA)/acrylic acid copolymer on the surface of polyethylene terephthalate (PET), which is considered to be a non-woven fiber, is also effective for the treatment of oil spills.[88] Chen et al.[89] reported a novel polymer resin namely poly(methyl methacrylate–co-pentaerythritol tetraacrylate-co-butyl methacrylate)-*g*-polystyrene for oil sorption by living radical polymerization on the surface of PS resin using ferric trichloride/iminodiacetic

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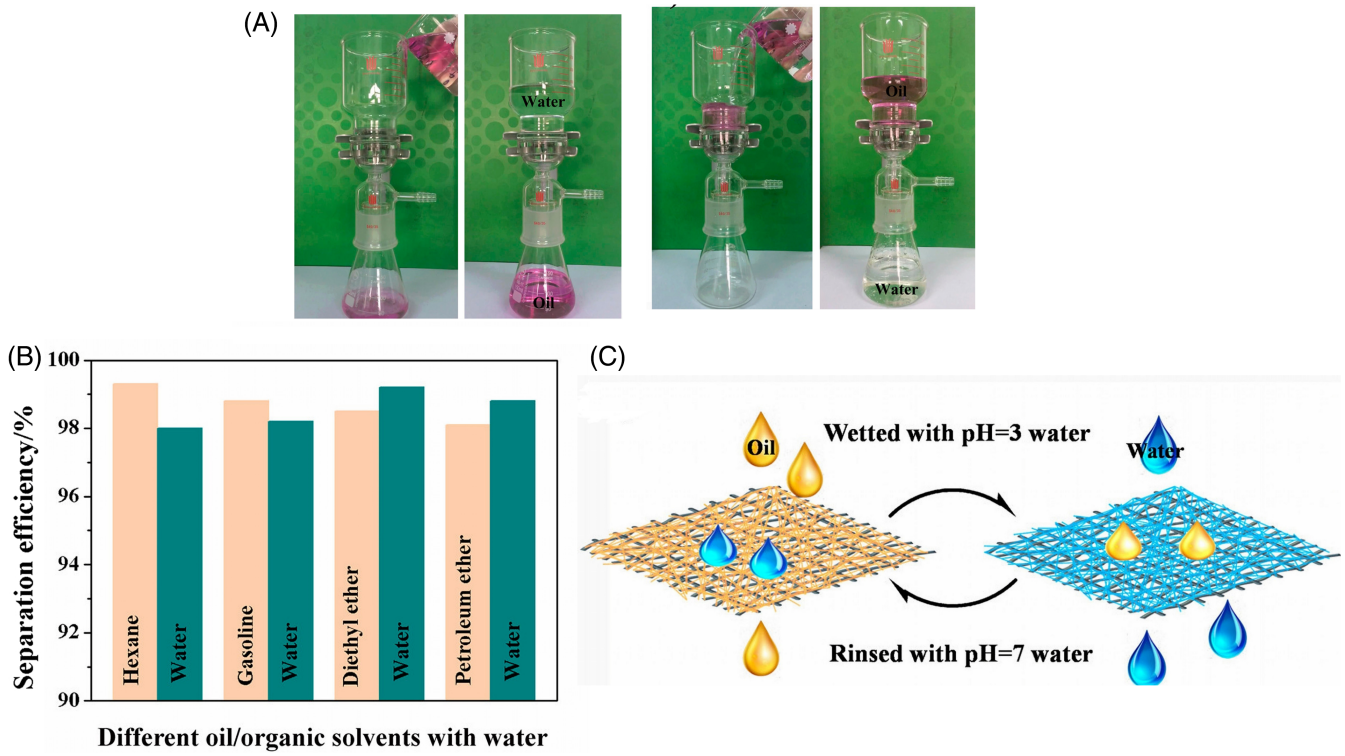

FIGURE 9 (A) Images showing pH-controllable oil/water separation device: Oil (*n*-hexane is colored with iodine) passed through the membrane, whereas the water was remained in the upper glass apparatus (right), if the membrane was wetted with acidic water (pH = 3) before separation, water could selectively permeate through the membrane, while oil could not (right). (B) Oil/organic solvent separation efficiency (dry state) and water separation efficiency (after wetted with pH 3 water) of four different mixtures using smart fiber membrane PMMA-*b*-P4VP/stainless steel mesh. The separation efficiency was determined by comparing the weight of oil or water before and after separation. (C) Schematic diagram of pH switchable smart fiber membrane PMMA-*b*-P4VP/stainless steel mesh. Reprinted with copyright permission from American Chemical Society[82]

acid as a catalyst. The oil sorption behavior of polymer was tested using toluene as the solvent. Results predicted that 21 g/g of polymer for toluene and the oil sorption process obeys the first order absorption kinetics. Kim and Jang[90] reported crosslinked styrene alkyl acrylate copolymer suitable for oil sorption. The various alkyl acrylate namely 2-ethyl hexyl acrylate, lauryl acrylate and stearyl methacrylate were used, which can be polymerized by suspension polymerization using benzyl peroxide as a cross linker. By altering the feed ratio of the prepared copolymer, the oil sorption properties can be altered. At a low concentration of acrylate, the copolymer did not interact well with the oil due to the lack of acrylate units with oil affinity. Similarly, at a higher concentration of acrylates, copolymers were fully swollen in oil without dissolving. This shows that the acrylate monomer appears a higher tendency toward branching and crosslinking than styrene does and is also directly proportional to the sorption capacity. Farag and Saeed[91] developed a docosanyl based copolymers using different monomer feed ratios of cinnamoyloxy ethyl methacrylate and methyl methacrylate in presence of a crosslinker. The hydrophobicity of the copolymer exhibited a swelling behavior and was used as an oil adsorber. The presence of long chain hydrophobic alkyl acrylate in the copolymer increases its solubility in crude oil. The oil absorbency of the copolymer in pure toluene exhibited 38 g/g, which shows maximum swelling capacity during few hours.

7 | INORGANIC POLYMER NANOCOMPOSITE SORBENTS

Inorganic polymer composites have advantageous over organic polymers due to their high sorption capacity, high surface area, and good selectivity. Khosravi and Azizian[92] reported polypyrrole/PU-based sorbent, which had a high tendency of oleophilic and hydrophobic characteristics for oil spill clean-up, and utilized a simple vapor-phase deposition followed by polymerization of polypyrrole and modification with palmitic acid (Figure 10A). The small quantity of Fe^{3+} introduced into the PU sponge plays a significant role in the establishment of polypyrrole. The prepared sponge could separate large amounts of various kinds of oil in the range of 22–62 g/g and the sorption process is very fast, which completes within 30 s. The dynamic sorption process was reported for most of the oils especially carbon tetrachloride, *n*-decane, *n*-heptane, and sunflower oil. The interconnected structure and macroscopic appearance were retained for the sponge

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even after 100 cycles (Figure 10B). The recyclability of the reported sponge can be harvested by squeezing the sponge under compression, which recovered its original form and could be reused for enormous number of oil/water separation. Wu et al.[93] reported novel oil adsorbent, fabricated by modifying PU sponges with titanium dioxide (TiO_2) solution (Figure 11A). PU sponge is a 3D porous polymeric product with low density, high porosity, high absorption ability, and good elasticity, making it a good candidate as an adsorbent. The attachment of TiO_2 nanoparticles reduced the hydrophilicity of the modified PU sponge, significantly by increasing its surface roughness and changing the chemical composition of the surface. The modified PU sponges exhibited oil absorption capacity of 95–110 g/g and found to be reusable up to 12 cycles holding 70% of its initial uptake capacity (Figure 11B).

Similarly, Liu et al.[94] created a cost-effective reduced graphene oxide-coated polyurethane (rGPU) sponge for effective oil-spill remedy. Although, GO is hydrophilic, the reduction of GO changes the materials from hydrophilic to hydrophobic due to reduction in the polar groups. The super-hydrophobicity and super-oleophilicity properties of the rGPU sponge make it a real tool for good oil/water separation capacity. When a piece of rGPU sponge was forced to contact with chloroform/water system, the chloroform was quickly sucked

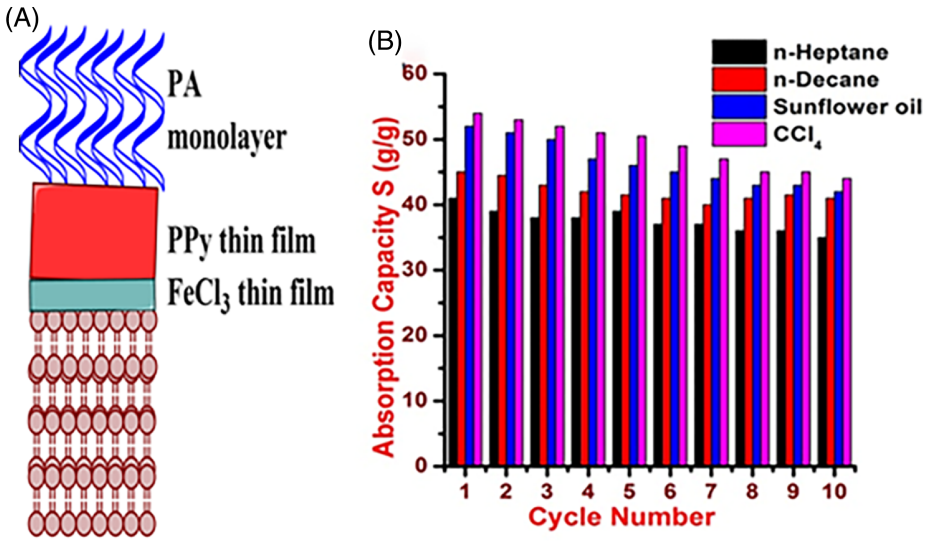

FIGURE 10 (A) Schematic representation of PPy-PA sponge. (B) Recyclability of the PPy-PA sponge for different oils times. Reprinted with copyright permission from American Chemical Society[92]

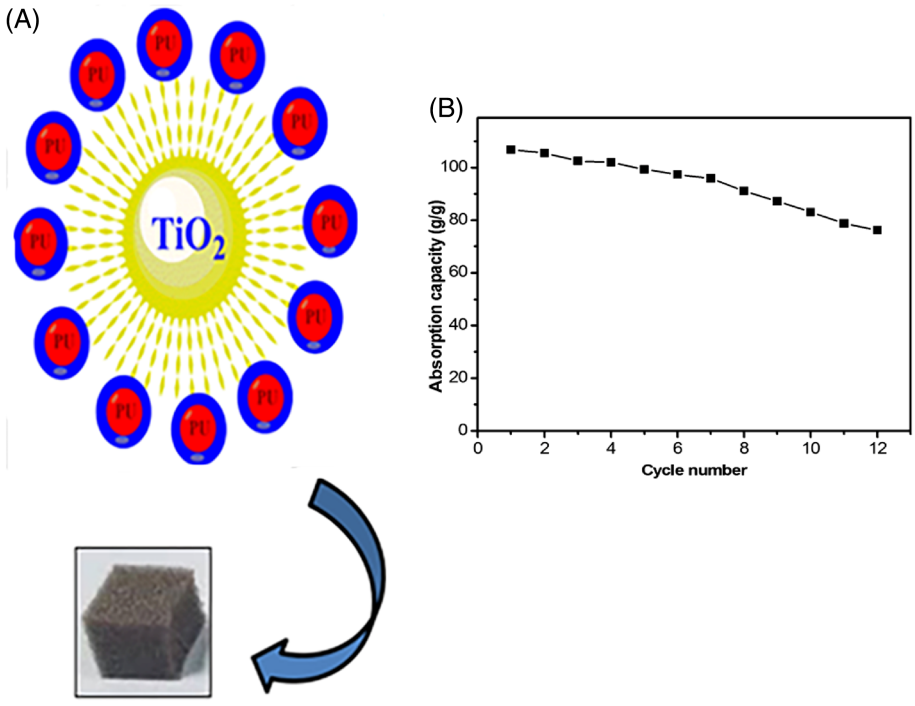

FIGURE 11 (A) Graphical representation of TiO_2/PU foam. (B) Oil absorption capacities of TiO_2/polyurethane in different cycles of absorption process. Reprinted with copyright permission from American Chemical Society[93]

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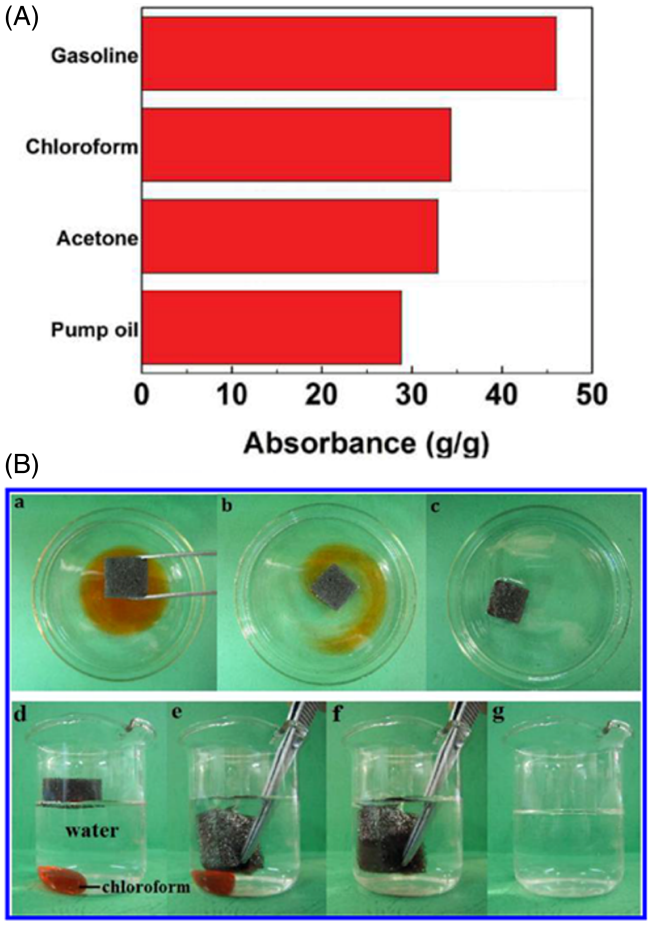

FIGURE 12 (A) Absorbance capacity of r-graphene oxide-polyurethane foam in various organic media. (B) (a–c) Selective absorption of diesel oil (dyed with Sudan I) on water by the rGPU sponge and (d–g) selective absorption of chloroform (dyed with Sudan III) in water by the rGPU sponge. rGPU, reduced graphene oxide-coated polyurethane Reprinted with copyright permission from American Chemical Society[94]

into the rGPU sponge within a few seconds. This makes the rGPU sponge a promising candidate for absorbing and eliminating high-density organic solvents in water. The adsorption rates of the rGPU sponge for all the oils are rapid (≤100 s) and are related to the viscosity of the oils (Figure 12A). The recyclability of an adsorbent and the recoverability of the absorbed oils and organic solvents are key requirements in practical oil cleanup applications. In this case, the absorbed diesel oil in the rGPU sponge can be simply recovered by manual squeeze (Figure 12B). Tjandra et al.[95] reported an oil adsorber by introducing reduced GO on the surface of PU in the presence of coupling agents. 3-Aminopropyl triethoxy silane and titanium triethanolaminatoisopropoxide were used as coupling agents and introduced before and after reduction of GO to compare its adsorbance capacities. The sponge coupled with titanium triethanolaminatoisopropoxide before the reduction of GO exhibits high oil absorbance in several organic solvents and reported to absorb 40–45 g/g (Figure 13). Foster et al.[96] reported iron oxide nanoparticle grafted polymer cluster for applicable in oil in water emulsions. The grafting methodology was claimed to synthesize oil in water cluster

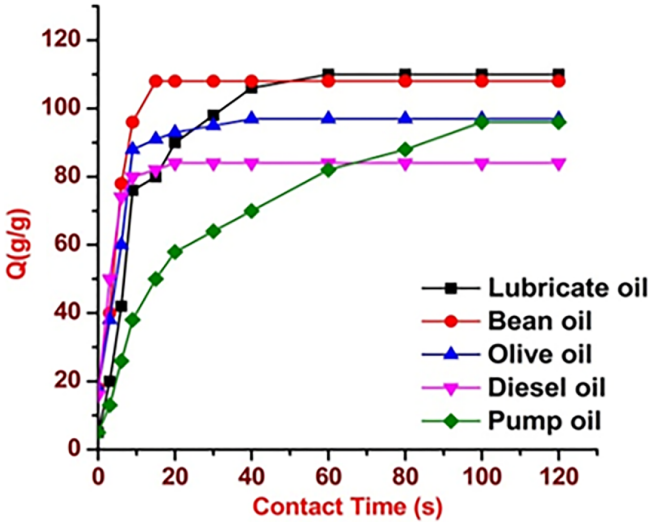

FIGURE 13 Absorption capacity of the rGPU sponges for various organic liquids. rGPU, reduced graphene oxide-coated polyurethane Reprinted with copyright permission from American Chemical Society[95]

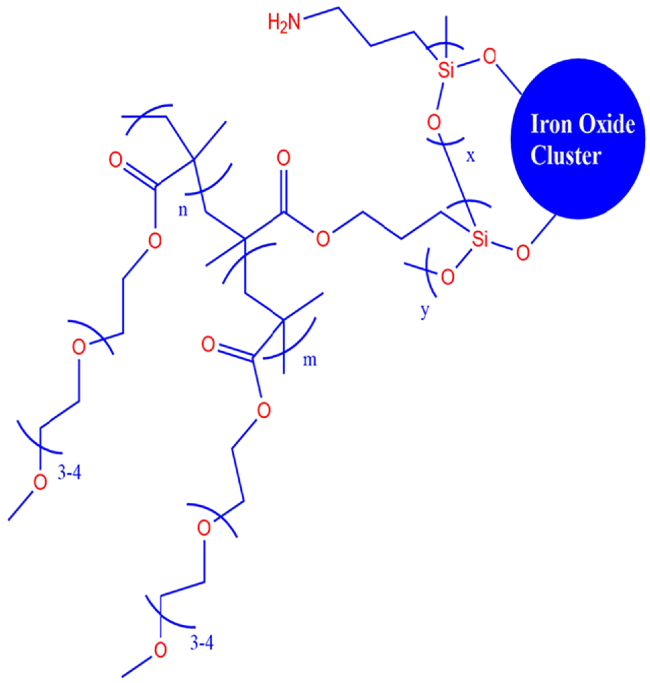

FIGURE 14 Chemical structure of "grafting through" of poly(oligo(ethylene oxide)monomethyl ether methacrylate) on 3-(trimethoxysilyl) propyl methacrylate-3-(aminopropyl) triethoxysilane functionalized iron oxide nanoclusters. Reprinted with copyright permission from American Chemical Society[96]

by using poly(oligo(ethylene oxide)monomethyl ether methacrylate (POEOMA) on 3-trimethoxy silyl propyl methacrylate-3-amino propyl triethoxy silane functionalized iron oxide nano clusters initiated by 4,4-azobis(4-cyanovaleric acid) to produce iron oxide POEOMA hybrids (Figure 14). This grafted iron oxide-polymer cluster exhibits a change in rheological properties and also has the impact on o/w

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emulsions. The synthesized polymer clusters have a higher entanglement in water, producing a viscosity up to 35,000 cP. The nano polymer clusters have enhanced ability to influence interfacial activity and rheology, which are in turn applicable to stabilization of emulsions and foams.

Liu et al.[97] reported a superhydrophilic and superoleophobicvpoly (sulfobetaine methacrylate) (PSBMA) grafted glass fiber filter for better oil water separation. The coating of zwitterionic polymer to the glass surface that can effectively separate oil and water (Figure 15A). PSBMA grafted glass fiber filter resulted in separating oil from water even without allowing miniscule amounts of visible oil to permeate through (Figure 15B). Pelletier and Siron[98] reported a silicon-based polymers as oil spill treatment agents. The oil treatment agents were prepared by reaching polyoxyethylenic surfactant, alkyl alcohols or carboxylic acids with alkyl chlorosilanes in light hydrocarbon solvents.

The treating solution undergoes rapid hydrolysis up on contact with the water surface and the reaction products can encapsulate and solidify oil slicks and other floating hydrocarbon. The recovery of the silicone can be performed by simple fish netting or by means of mechanical operation suitable for separating the floating semi-solid residue from water. Oil silicone residue netted from the water surface reaching 80%–85% of total weight. Wu et al.[99] reported graphene based PU foam by reducing the GO and gets assembled on the skeleton of PU foam to make suitable and more efficient to trap oil from oil/water mixture. Shuai et al.[100] demonstrated a poly(dimethylsiloxane)—titanium dioxide (PDMS-TiO_2) composite coated on PU sponge, which shows to achieve superhydrophobicity with a water contact angle of 154° and more effectively applied for water/oil separation. Similar results were also proved by Choi et al.[101] and Kim et al.[102] by utilizing PDMS and PDMS-TiO_2 sponge with high elasticity and hydrophobicity-superoleophilicity in air and oleophobicity under water in UV irradiation. Han and Steckl[103] produced a core sheath structured micro/nano fiber comprising of Teflon sheath and poly(caprolactone) by electrospinning. This material has a high degree of superhydrophobic/oleophobic properties and produces a new route to choose material for creating superoleophobic surfaces. Xue et al.[104] reported a super oleophobic hydrogel coated mesh to treat oil/water separation, such as vegetable oil, gasoline diesel, and crude oil, without any extra power. Cao et al.[105] modified a PDA coated stainless steel mesh with *n*-dodecyl mercaptan through organic mediated Michael addition reaction resulting in an excellent hydrophobic and superoleophilic properties to treat oil/water separation. Zhang et al.[106] modified the surfaces of PU sponge with fluoroalkyl silane to treat oil/water mixture, which produced a water contact angle of 155° for effective separation. Similarly, PU sponge was modified with methyltrichlorosilane to tune their absorption properties, which produced a good recyclability up to 400 cycles. Ono et al.[107] reported a polyelectrolyte gel comprising of poly(octadecyl acrylate) and tetraalkylammonium tetrakis (3,5-bis(trifluoromethyl)phenyl borate to investigate the swelling behavior of organic solvents and oils. Further, they reported that polyelectrolyte has an uptake capacity, which is 300 times much higher than its dry weight. Another type of polyelectrolyte gel was reported by Pourjavadi et al.[108] by comprising of vinyl imidazoline and dodecyl methacrylate. The gel has a good swelling capacity nearly 50 times than their dry weight in organic solvents. Li et al.[109] reported a nanoporous polydivinylbenzene (PDVB)-PDMS decorated melamine sponge for facile oil–water separation. Compared with most of the oil adsorbents, the reported melamine sponge shows oil-sorption capacity in the range of 40–120 g/g for most oils/organic, which immediately collects large-scale oils in situ from water surface. Similarly, Wang et al.[110] reported a novel luffa sponge modified by a polyhedral oligomeric silsesquioxane-based compound, which could essentially take up all the components of the oil during the absorption process. Wu et al.[111] reported a perfluorooctyltriethoxysilane functionalized CNTs in PS matrix and electrospun to create an effective and rigid sorbent for oil. The reported sorbent could be a cost-effective, convenient, and highly efficient oil sorption material for oil spill cleanup at large production scales.

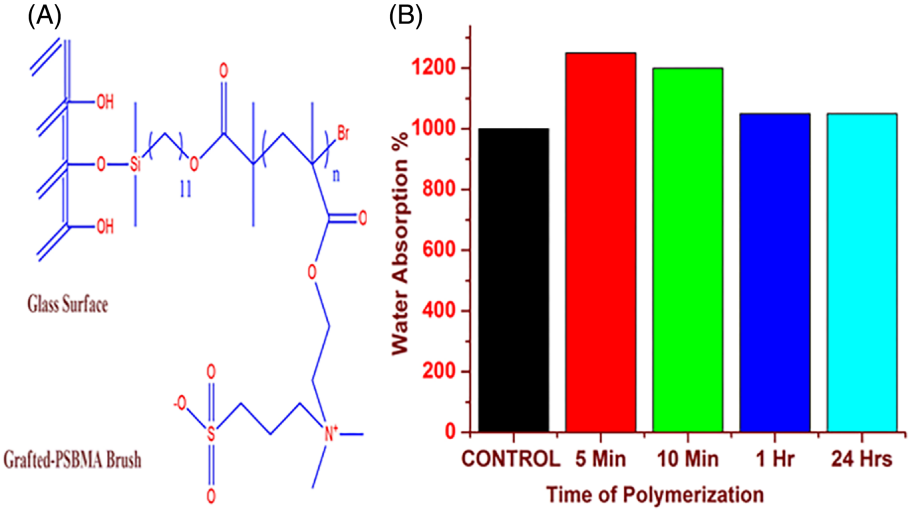

FIGURE 15 (A) Chemical structure of grafting of PSBMA on glass surfaces via surface initiated ATRP. (B) Water absorption of the control glass fiber filter and the poly(sulfobetaine methacrylate) grafted glass fiber filter with different grafting times. ATRP, atom transfer radical polymerization; PSBMA poly(sulfobetaine methacrylate) Reprinted with copyright permission from American Chemical Society[97]

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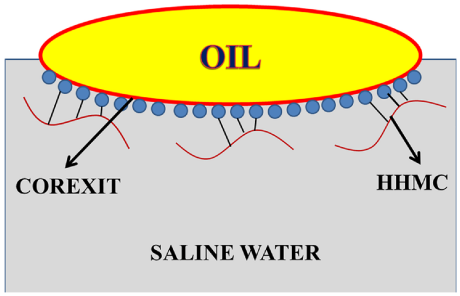

FIGURE 16 Scheme showing crude oil corralled by corexit and bound by HHMC in a gel-like aggregate hydrophobically modified chitosan with corexit. Reprinted with copyright permission from American Chemical Society[126]

8 | BIOPOLYMER NANOCOMPOSITE SORBENTS

Apart from artificial/synthetic absorbers, natural absorbers are used nowadays for environmental issues to conserve both environmental and economic aspects.[112] A wide range of natural sorbents, such as rice straw,[113] corn corb,[114] peat moss wood,[115] cotton,[116] milkweed floss,[117] kapok,[118] kenaf,[119] and wool fibers,[1] have been employed largely in oil spill cleanup. Biopolymer composites are more prone to environmental remedies especially tracing oil spills and toxic metal ions. They are biodegradable, environmentally friendly, non-toxic, and easier complexation with the contaminant. Even though, it has some disadvantages, such as low efficiency, these sorbents have high thermal stability, oleophilicity, hydrophobicity, and recyclability of oil. The hydrophilic property of the sorbent allows the oil to be kept on the surface of the material due to the adhesive force and the active functional groups, which determines its high sorption and ion-exchange capacity. In addition, its cost effective and environmentally friendly nature makes it a convenient tool for oil-contaminated clean up. Olga et al.[120] investigated the oil adsorption process using three natural plant origin adsorbents, namely a peat moss (sphagnum), corby, and spillsorb. They also found that acid treated peat moss carbonized at the temperature of 200–250°C and modified by acetic acid to gain high sorption capacity compared to others. The peat moss has higher efficiency, leaving residual oil content less than 0.03 g/L. These sorbents have high buoyancy, oil capacity, and low sedimentation velocity. The sorbents are porous in nature, making them to absorb the oil products well and retain them for a long time. Athas et al.[121] reported the ability of natural amphiphiles to disperse crude oil in seawater. They utilized lecithin, extracted from soybeans, and Tween 80, a surfactant. These amphiphiles have a good synergistic effect and pack closely at the oil–water interface. In addition, it has low desorption and high steric repulsions between the oil droplets, which prevents their coalescence for a much longer time and potentially be a viable alternative for the dispersion of oil spills. Choi and Cloud[122] reported the significant sorption capacity of Milkweed and Cotton fiber to trace the crude oil from the surface

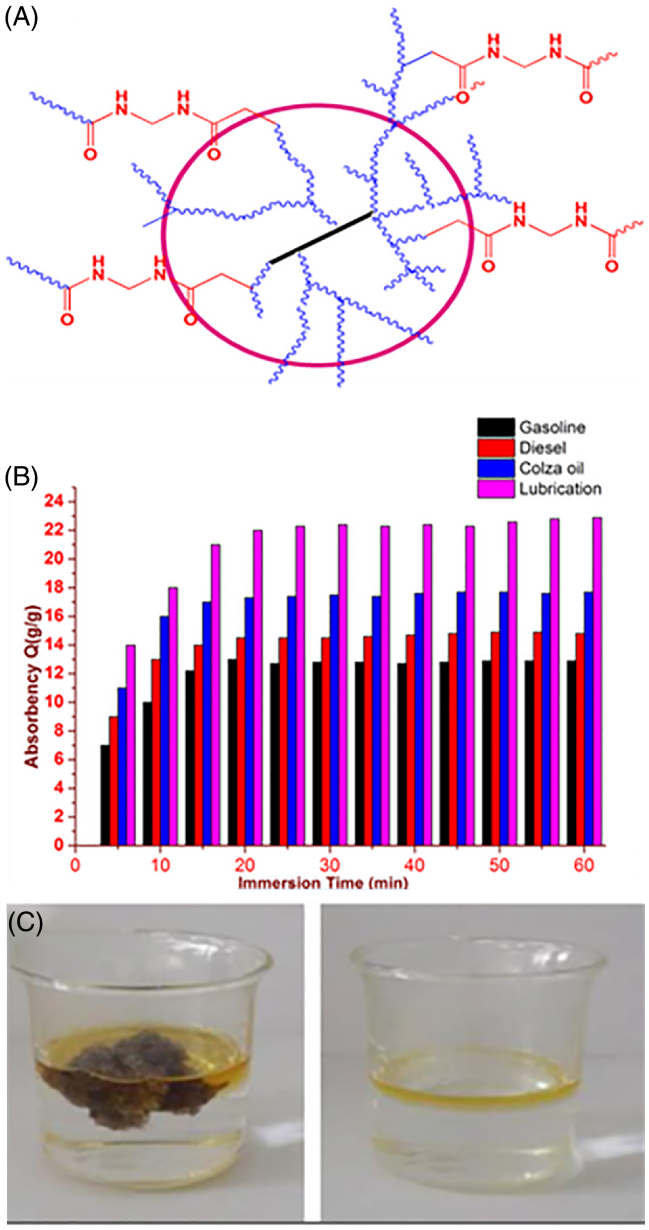

FIGURE 17 (A) Graphical representation of NSM-*g*-P(MMA-*co*-BA) composite resin network. (B) Oil absorbency of NSM-*g*-P(MMA-*co*-BA) resin in mixture of 0.9 wt% NaCl solutions and oil as a function of immersion time. (C) Removal of colza oil from 1 wt% NaCl solution using NSM-*g*-(PMMA-co-BA) Reprinted with copyright permission from Wiley online library[127]

of sea water. The high oil sorption capacity of Milkweed fiber was due to the large amount of wax and non-collapsing lumen on the fiber surface, which absorbs nearly 40 g/g of polymer. They demonstrated that sorbents could be recovered and recycled for oil spill clean by using simple mechanical retrieval equipment. Thus, natural sorbent materials could be beneficial in the oil spill cleanup in full or partial substitution of synthetic sorbents. Kavalenka et al.[123] demonstrated a superhydrophobic micro haired wood-based surface for both water and oil removal in a mixture. Bazargan et al.[124] recently reported a naturally occurring rice husks

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as an oil sorbent material due to its low bulk density and fluffy structure. Sun et al.[125] recently reported the synthesis of rice straw acetates for effective oil spill clean ups. Oil sorption capacities of nearly 16.8–24.0 g/g for acetylated straw have high sorption capacity, high uptake rate, ease of desorption, convenient recovery and recyclable for oil spills. All the literature reports discussed that they clearly imply the advantages of using natural sorbents for oil spill cleanup, including its properties like biodegradability and economically acceptable nature. However, it was also observed to have some disadvantages, namely poor buoyancy characteristics and low hydrophobicity, which makes it to absorb water quickly and sink when deployed in the sea. Oil sorption capacity of these natural sorbents can increase further by blending it with other synthetic products in an optimized ratio, which renders high hydrophobic and buoyancy characteristics. John and coworkers[126] proved that a modification of biopolymer also served as a better oil adsorber. They utilized a hydrophobically modified chitosan, which was considered to be nontoxic and biocompatible that enhanced the stability of the crude oil droplets dispersed in saline water as shown in Figure 16. They utilized a combination of biopolymer and the chemical dispersant, namely corexit, to remain stable for extended period. The hydrophobic residues from the polymer preferentially anchor in the oil droplets form a layer of the polymer around the droplets, which increase the colloidal stability. The authors also revealed that the combination of biopolymer and chemical dispersant potentially reduced the use of dispersants. Further, increasing the molecular weight of the polymer would change the rheological properties of the oil in water emulsions to that of a weak gel.

Hai et al.[127] reported a novel polymer composite resin material to treat oil spills. They attempted to synthesize a lipophilic composite resin by grafting copolymerization of methyl methacrylate and butyl acrylate on the surface of nitraria seeds meal (NSM), an important economic crop belonging to the zygophyllaceous (Figure 17A). The oil sorption capacity of the polymer composite resin was tested with different oils, namely gasoline, lubrication, colza, and diesel oil, respectively. The maximum sorption capacities of the polymer composite resin, namely NSM-*g*-P(MMA-co-BA), is up to 29.6 times of their weight for lubrication, 24.3 times for colza oil, 22.7 times for diesel, and 21 times for gasoline (Figure 17B). The absorbed oil can be recovered easily by simple extraction and hence considered to be a promising substitute for the conventional adsorbent material used to treat large oil spills (Figure 17C). Nguyen et al.[128] illustrated a low density and highly porous cellulose aerogel for crude oil spill remedy. The methyltrimethoxysilane (MTMS) functionalized aerogel showed good hydrophobicity and oleophilicity with water contact angles greater than 140°. The aerogel exhibited high absorption capacities of 18.4, 18.5, and 20.5 g/g for three different crude oils, namely Ruby (RB), Te Giac Trang (TGT), and Rang Dong (RD). The author also reported that the viscosity of the crude oils is the main factor affecting their absorption onto the aerogel. The strong affinity of the MTMS-coated recycled cellulose aerogel to the oils makes the aerogel good adsorbent for targeting oil spills.[129]

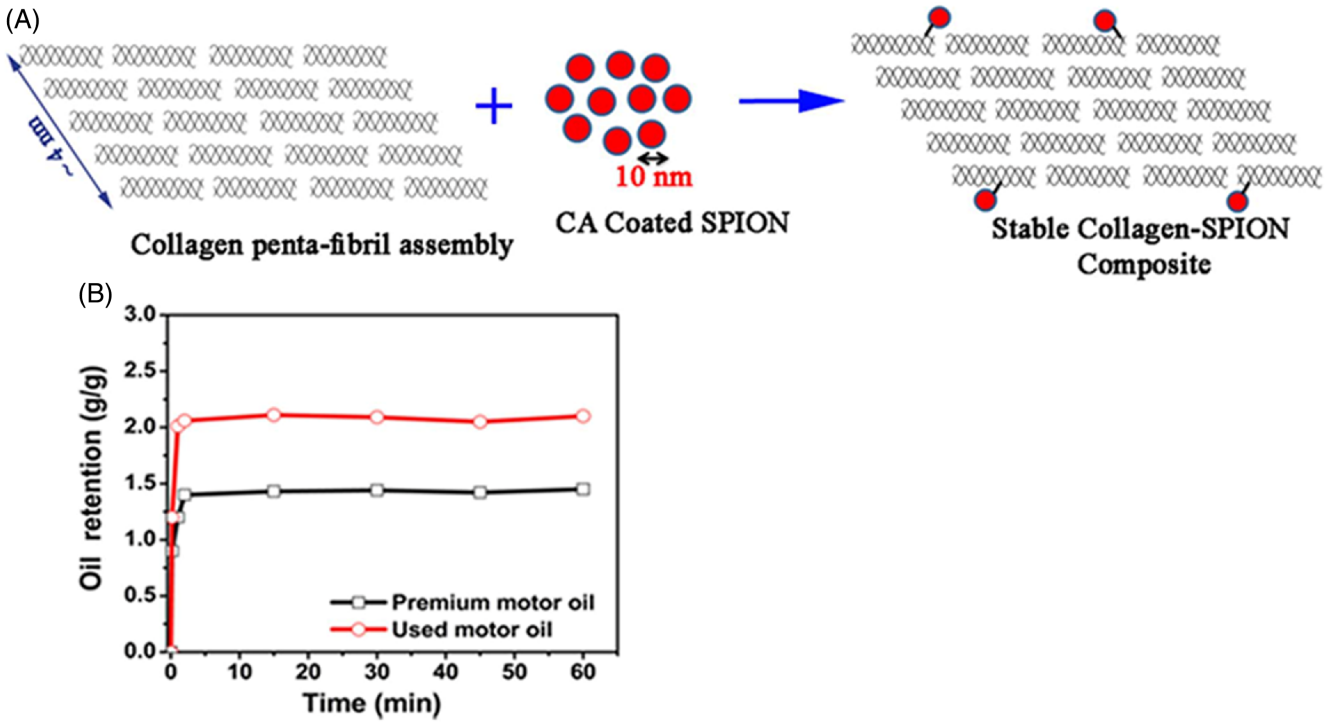

FIGURE 18 (A) Schematic illustration of collagen fiber SPION nanobiocomposite. (B) Oil retention ability of the collagen—SPION nanobiocomposite as a function of time using premium and used motor oil. SPION, superparamagnetic iron oxide nanoparticles Reprinted with copyright permission from Nature Publishing[132]

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A polydopamine-silicon dioxide nanocomposite was reported recently by Wang et al.,[130] which attains its super hydrophobicity even after 90 cycles of oil/water separation. In addition, it shows a good chemical stability even in high acidic and alkali medium. Similarly, Ruan et al.[131] reported a superhydrophobic crosslinked polydopamine coated commercial sponge for oil/water mixture, where a thin layer of polydopamine was self-polymerized on the surface of the sponge. Thanikaivelan et al.[132] reported the collagen polymer based magnetic nanocomposites for effective removal of oil. A stable magnetic nanocomposite of collagen and superparamagnetic iron oxide nanoparticles (SPIONs) (Figure 18A) coated over citric acid was simply prepared by utilizing protein wastes as demonstrated by the authors. SPION stabilizes the collagen waste fibers and further converts the oil-absorbed nanocomposite into a magnetic graphitic carbon material, which paves a new opportunity for converting bio-wastes into useful nanomaterials. (Figure 18B). Further the author claimed the recycling process by simple squeezing to remove the oils and the material used for repeated oil adsorption.

Elanchezhiyan et al.[133] reported the usage of hybrid composite materials to evaluate the equilibrium adsorption capacity for the removal of oil from oil in water emulsions. They utilized two types of hybrid composite materials made from biopolymers, which were chitosan/lanthanum–zirconium mixed oxide and chitosan/aluminum-zirconium mixed oxide. The sorption capacity significantly altered by pH because it played a vital role in oil sorption process and a maximum oil removal of 92% and 74% were achieved using these two composites (Figure 19). Nascimento and coworkers[134] evaluated the adsorption capacity of crude oil using biopolymer and derivatives comprising of chitin flakes, chitin powder, chitosan flakes, and chitosan powder. Chitosan flakes presented superior adsorption of oil, which forms a polymeric film on oil slick to restrain and remove the oils from water. The hydrothermal and acetylation treatment of the biopolymer populus fiber recently had increased attention in separating oil water mixture. The thermal treatment of the biopolymer fiber results in a lignocellulose, which is a new class of crosslinked polymer with higher efficiency for oil sorption. The maximum sorption capacity of hydrothermally treated populus fiber is due to thermal stability and

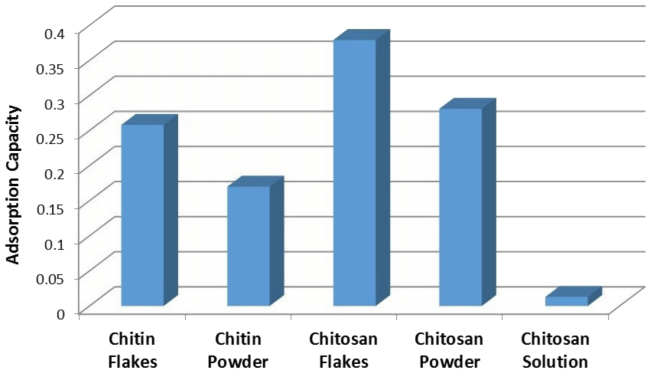

FIGURE 20 Adsorption capacity of the oil using chitin and chitosan based biopolymers. Reprinted with copyright permission from publons55[134]

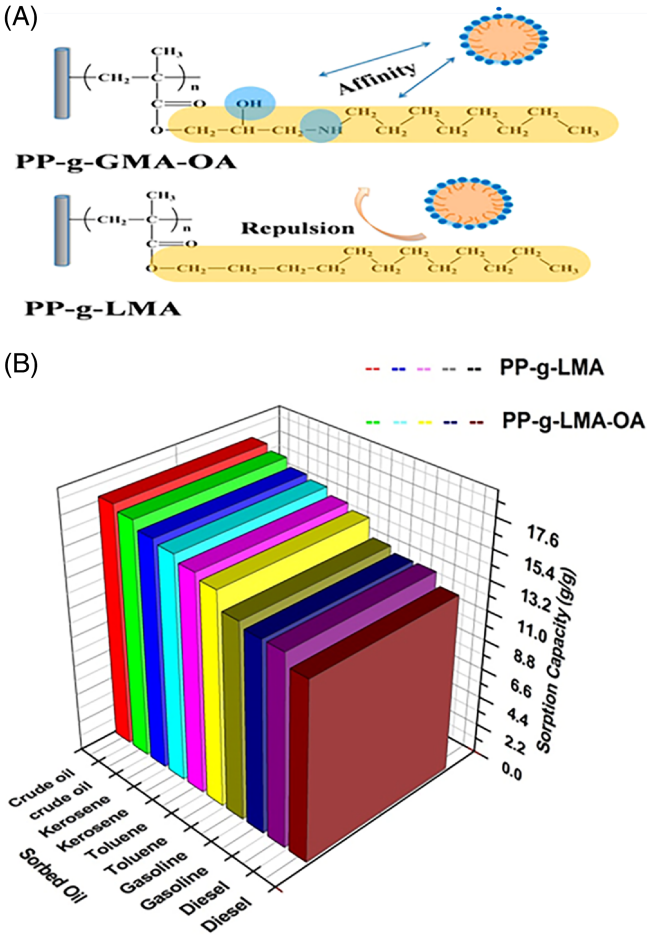

FIGURE 21 (A) Chemical illustration of PP-*g*-GMA-OA and PP-*g*-LMA nonwoven fiber. (B) Sorption capacity of PP-*g*-LMA and PP-*g*-GMA-OA in organic solvents/oils Reprinted with copyright permission from American Chemical Society[138]

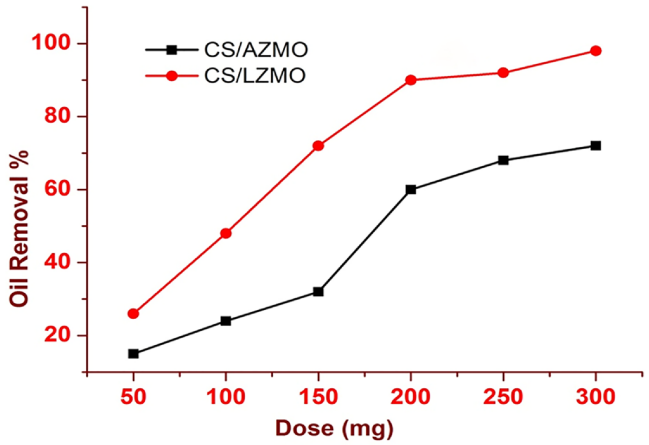

FIGURE 19 Oil removal percentage of dosage chitosan/lanthanum–ziconium mixed oxide (CS/AZMO) and chitosan/aluminum–zirconium mixed oxide (CS/LZMO) at pH 3. Reprinted with copyright permission from Wiley online library[133]

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the lumen structure of the polymer to maximize oil entrapment (Figure 20).

9 | MODULATION OF POLYMERIC ARCHITECTURE FOR OIL SORPTION

Various polymers and copolymers were prepared with different architectures and successfully tested for oil sorption performance through emulsion,[135] suspension,[136] and radical polymerization.[137] Zhou et al.[138] reported a fabrication of hydrophobic and hydrophilic sites on polypropylene (PP) non-woven for oil spills remedy. The polypropylene nonwoven with hydrophilic and hydrophobic sites was prepared by grafting glycidyl methacrylate (GMA) onto PP nonwoven through graft polymerization and subsequently converted the epoxy group of GMA to hydrophilic hydroxyl and hydrophobic long-chain alkylamine through reaction with *n*-octylamine (Figure 21A). For oil in-water emulsion, the hydrophilic end of surfactant faces outward,

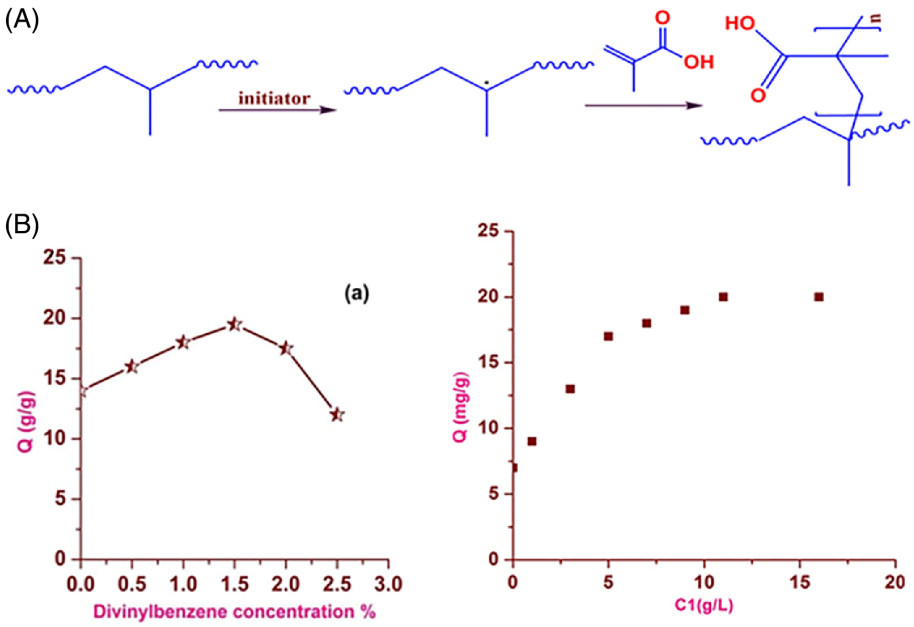

FIGURE 22 (A) The grafting process of PMAA onto PP backbone. PP-*g*-PMAA fibers. (B) Influence of DVB concentration to the oil sorption capacity of PP-*g*-MAA. DVB, divinylbenzene Reprinted with copyright permission from OMICS International[139]

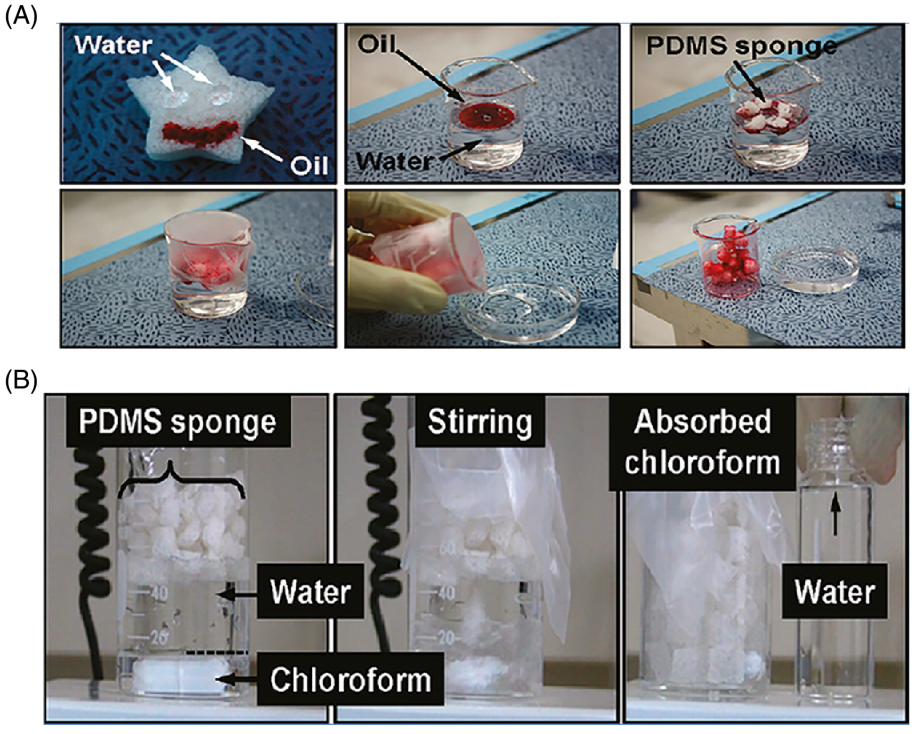

FIGURE 23 (A) Photographs showing hydrophobic and oleophilic properties of PDMS sponge, a layer of oil (dyed with oil Red O) was removed by the sponge to the oil. (b) Absorption process of the PDMS sponge in a mixture of chloroform and water. PDMS, polydimethylsiloxane Reprinted with copyright permission from American Chemical Society[101]

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which lowers the interaction between the hydrophilic interface and hydrophobic surface.

The hydrophobic site preserved its hydrophobicity and sorption capacity, and the hydrophilic site on PP surface effectively increased the affinity between the hydrophilic interface of emulsion and sorbent. The overlapped and intertwined structures could provide spaces large enough to accommodate oil and emulsion. Compared with hydrophobic PP-*g*-LMA nonwoven, merely a small decrease could be observed and amphiphilic PP-*g*-GMA-OA nonwoven showed a moderate sorption capacity (Figure 21B). This is mainly ascribed to the inherent hydrophobicity of polypropylene substrate (hydrophobic carbon backbone) and the introduction of hydrophobic site.

Ha et al.[139] reported polypropylene-*g*-methacrylic acid oil sorption materials by graft polymerization. Methacrylic acid was grown on to the polypropylene fibers using divinylbenzene (DVB) as crosslinking agent in presence of AIBN (Figure 22A). The oil sorption capacity of pristine PP fiber can be improved by suitable modifications via graft polymerization to alter its chemical properties and sorbent surface. It is also reasonable to believe that the PP fiber hydrophobic–oleophilic characteristics could be attributed a waxy surface, leading to excellent oil absorbency and retention efficiency. A strong rising concentration of DVB causes a decreased oil sorption capacity of the material due to the critical increase of crosslinking level. The oil sorption capacity of PP-*g*-MAA fibers was revealed a maximum oil sorption capacity of 20.833 g/g (Figure 22B). Choi et al.[101] present a sugar-template PDMS sponge for the selective absorption of oil from water. The proposed PDMS sponge can be elastically deformed into any shape, and it can be compressed repeatedly in air or liquids without collapsing. Therefore, absorbed oils and organic solvents can be readily removed and reused by simply squeeze (Figure 23A). The PDMS sponge also showed high absorption capacity not only for nonpolar organic solvents but also for polar organic solvents. For instance, the absorption capacity of 1,2-dichlorobenzene (a well-known toxic organic contaminant in water treatment) for the PDMS sponge was approximately 1000%.

Since, the PDMS materials show a relatively excellent chemical inertness, PDMS sponge may possibly be used for both nonpolar and polar organic solvents, with great potential for the removal of toxic organic contaminants and oil spills from water (Figure 23B). Further, the similar work was reported by Zhang et al.[140] by a modified sugar template with a three-dimension interconnected pores and a swellable skeleton by curing the prepolymer in a p-xylene solution in the presence of commercial sugar particles (Figure 24). The synthesized material is porous and organo-gelly that can have properties of both super hydrophobicity and superoleophilicity. Unlike other porous oil adsorbents, the obtained PDMS oil adsorbents are swellable, which can increase their absorption capacities further. Absorption capacities of PDMS oil adsorbents are 4–34 g/g for various oils and organic

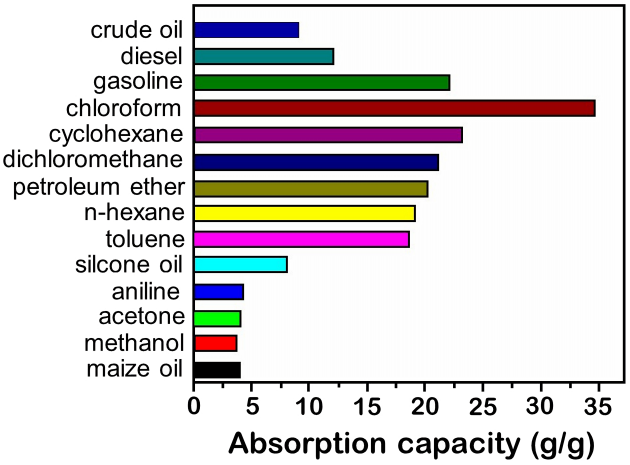

FIGURE 24 Mass adsorption capacities of the 3D PDMS oil adsorbents for various organic solvents. PDMS, polydimethylsiloxane Reprinted with copyright permission from American Chemical Society[140]

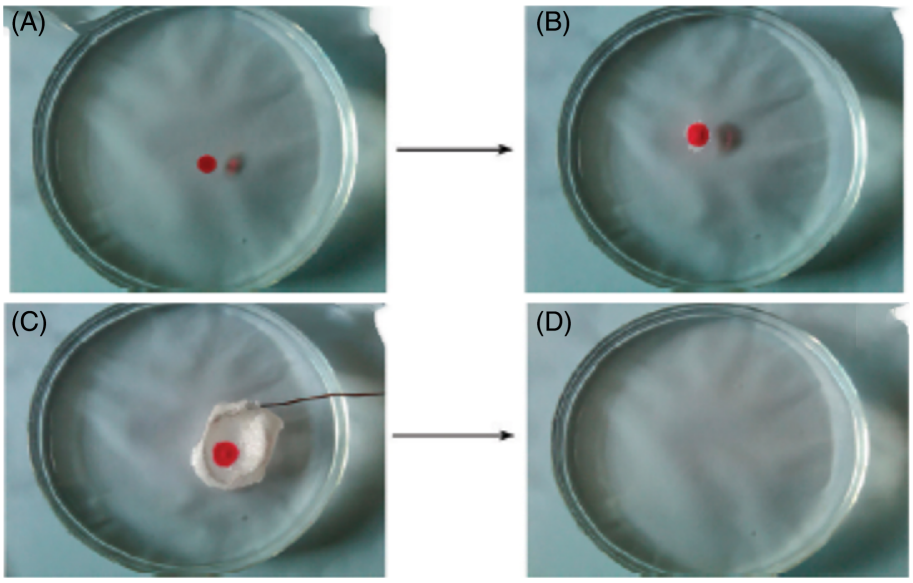

FIGURE 25 Photographs showing removal of motor oil from water surface using porous polysulfone microspheres. (A) Motor oil labeled by dye on the water surface. (B) PSF on the surface of oil over the water. (C) The oil saturated microspheres were moved away using homemade net. (D) Oil free water was obtained. PSF, polysulfone Reprinted with copyright permission from American Chemical Society[141]

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solvents, which are three times that reported previously. The absorption process can be finished within tens of seconds. Furthermore, the absorbed oils or organic solvents can be recovered by compressing the oil adsorbents, and after 20 absorbing/recovering cycles. Yu et al.[141] reported the monodisperse porous polysulfone (PSF) microspheres with hollow core/porous shell structure by a water-in-oil-in-water (W/O/W) emulsion solvent evaporation method. The prepared microspheres are developed as sorbents to remove oil from water due to their highly hydrophobic and super oleophilic properties. The PSF microspheres synthesized in the presence of oleic acid exhibit the best separation efficiency due to higher buoyancy (Figure 25). The microspheres with appropriate size, unsinkable properties, and excellent reproducibility can be quickly distributed and collected in seconds on the surface of water.

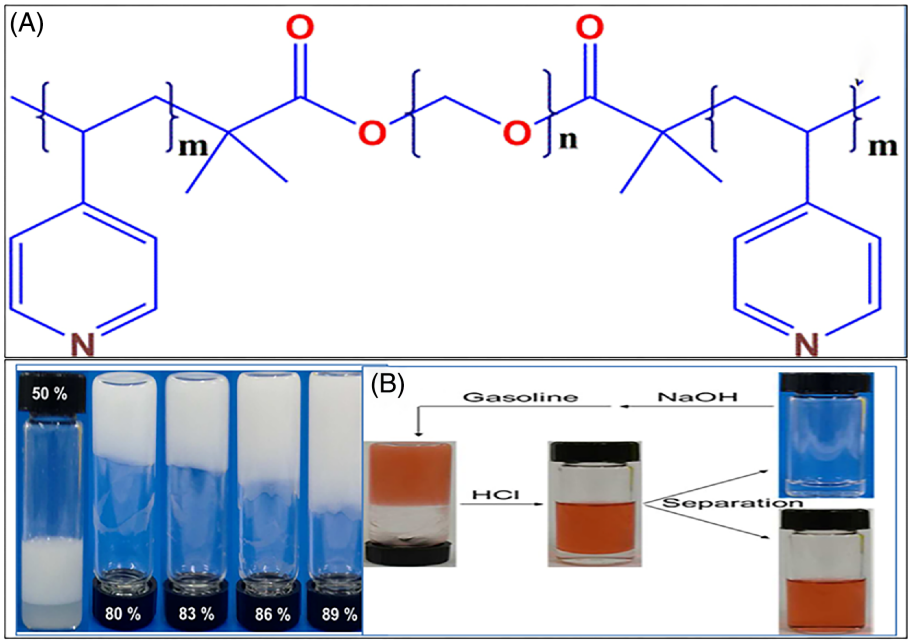

FIGURE 26 (A) Chemical structure of assembled block copolymer stabilized high internal phase emulsion hydrogels. (B) Photos of emulsion and HIPE hydrogels at different percentage of toluene and assembled BCP stabilized emulsions. BCP, block copolymer; HIPE, high internal phase emulsion. Reprinted with copyright permission from American Chemical Society[142]

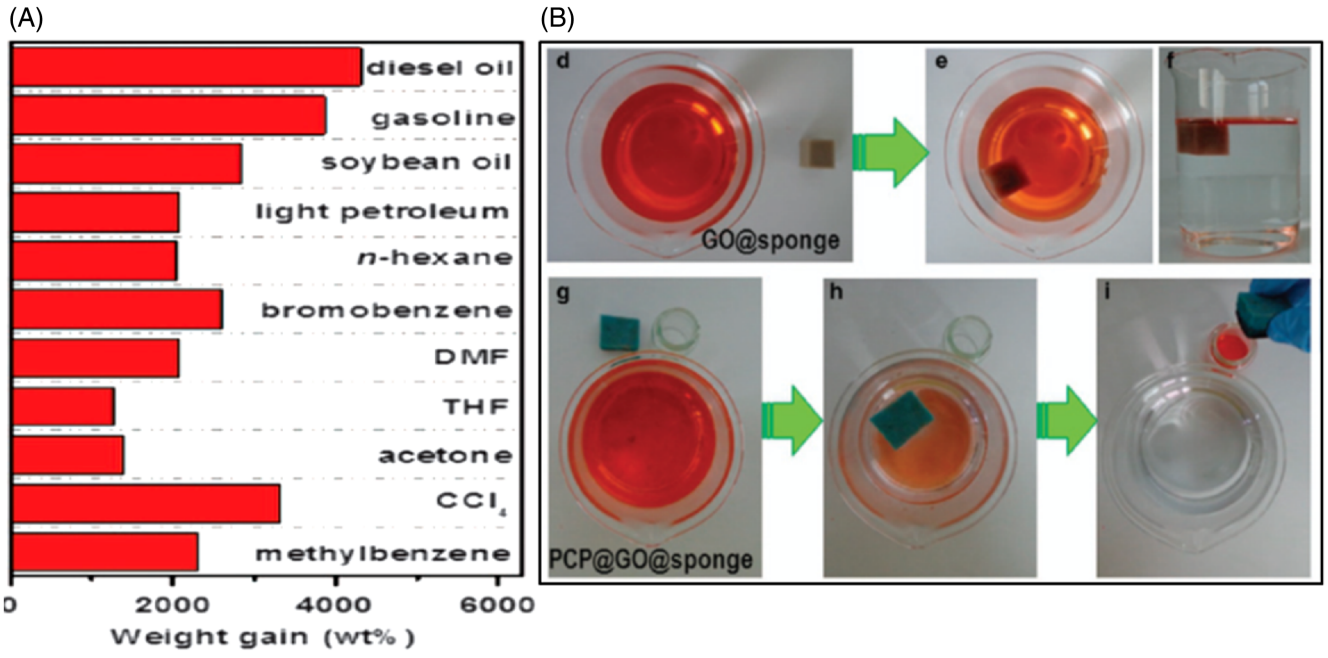

FIGURE 27 (A) The adsorption efficiency of porous coordination polymer @GO sponge for diverse oils and organic solvents. (B) Photographs showing the nonselective adsorption behavior of PCP@GO@sponge for (dyed red) gasoline/water and subsequent squeezing to recover the gasoline. Reprinted with copyright permission from Nature Publishing[143]

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Zhang et al.[142] reported a novel route to enhance oil safety by transforming oils into high internal phase emulsion (HIPE) hydrogels. These HIPE hydrogels are stabilized by assembled BCP, namely poly (4-vinylpyridine)-block-poly(ethylene glycol)-block-poly(4-vinylpyridine) (4VPm-EGn-4VPm) (Figure 26A), which shows oil absorption capacity as high as 89% with a variety of oils, including toluene, xylene, blended vegetable oil, canola oil, gasoline, diesel, and engine oil (Figure 26B). These assembled BCP stabilized HIPE hydrogels offer a novel and safe approach to enhance oil safety and to prevent environmental damage. Jiang et al.[143] reported a novel and highly stable in-situ porous coordination polymer@grapheneoxide@sponge with a corrugated carbon trifluoride surface (PCP@GO@sponge) that features high hydrophobicity for the remediation of oil spills and organic solvent leakage into the water. The polymer post-synthetic modification with fluoro-alkyl groups occupies the closest F–F distance between adjacent layers on the surface. The sorbentPCP@GO@sponge promotes fair stability and generates a narrow hydrophobic interlayer space that fully prevents the entry of water molecules and takes up as much as 1200–4300 wt% of organic pollutants and oils (Figure 27A). In addition, the sorbent is even able to possess a similar adsorption capacity after being immersed in different pH solutions for 24 h. Overall PCP@GO@sponge are reported to be an excellent sorbent with good hydrophobicity/oleophilicity and its facile fabrication and continuous operation without a capacity limit (Figure 27B). Brostow et al.[144] synthesized a modified polyethylene terephthalate fiber from recycled PET bottles by crosslinked polymer coatings to the fiber surface (Figure 28). The surface was coated with OA, maleic anhydride (MA), and related esters of methyl acrylate based polymers and copolymers via suspension and bulk polymerization techniques using DVB as the crosslinking agent. The material shows successful efficiency as an oil adsorber with a capacity of 90.2 g/g of polymer in 60 min. This phenomenon may be explained by considering that oil absorption by composites is probably synergized by the polymer coatings and the nonwoven PET supporter. The formation of elastic and porous network based on ODA and MA apparently provides large specific surface areas as well as large interstitial spaces.

Chung and Yuan reported a novel solution to oil spill recovery using thermo degradable polyolefin oil super adsorbent polymer by a crosslinked polyolefin terpolymer containing styrene and DVB units exhibiting both oleophilic and hydrophobic units with the hydrocarbons in crude oil (Figure 29A).[70] This crosslinked structure shows rapid oil adsorption and swelling to reach a capacity 45 times of their weight (Figure 29B). The capacity of oil uptake is inversely proportional to the crosslinking density. The recovered oil-swelled gel, containing more than 98% oil and 2% crosslinked polyolefin terpolymer, is suitable for regular oil-refining processes (Figure 29C). Pinto et al.[145] reported pristine PU foams with highly interconnected open porous structures with high oil absorption capacity. They claimed that chemical functionalization of the porous structure does not enhance the oil absorption efficiency, but it significantly contributes to the increase of the process selectivity. Oil absorption capacities higher than 40 v/v (or 30 w/w) were obtained using pristine PU foams. Open

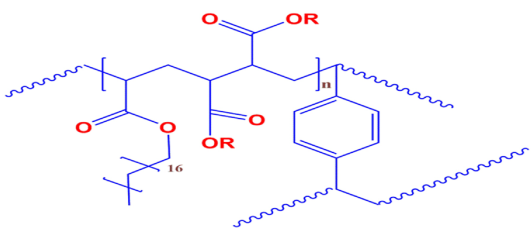

FIGURE 28 Chemical structure of nonwoven poly ethylene terephthalate (NWPET) fiber. Reprinted with copyright permission from RSC Publishing[144]

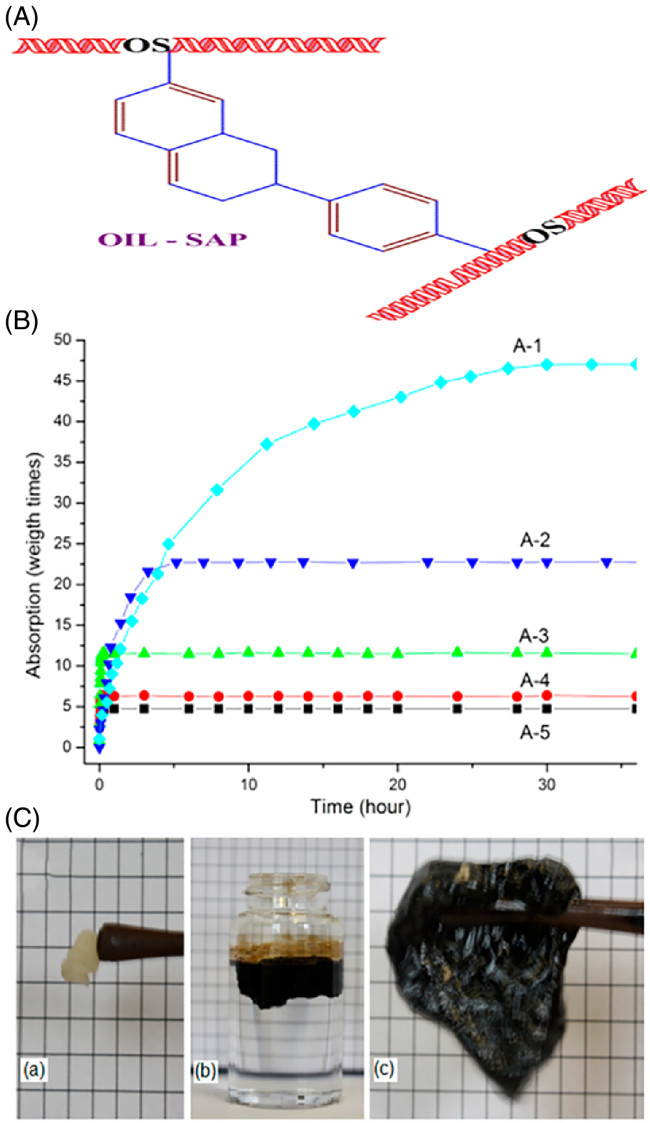

FIGURE 29 (A) Graphical illustration of crosslinked x–OS—DVB terpolymer material for oil absorption—thermodegradable polyolefin oil superabsorbent polymer. (B) Oil uptake versus time for a comparative set of x–OS—DVB samples (runs A-1–A-5) with a crude containing approximately 70% volatile light oils and 30% nonvolatile heavy oils. (C) Photographs showing the removal of the crude oil using cross linked poly olefin super absorbent polymer (OS-DVB-oil SAP). Reprinted with copyright permission from American Chemical Society[70]

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TABLE 3 Comparison of various polymeric oil adsorbents

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
1.	Urethane-isocyanate alcohol polymer	Motor oil	Pre-reacting a poly isocyanate with a long-chain monohydric alcohol to form a prepolymer. Following the formation of this prepolymer, a polyol is combined resulting in development of a cellular polyurethane	Granular	35	NA	Low	Good degree of elasticity and excellent thermal stability	148
2.	Acrylate-nitrile alcohol polymer	Crude oil	Copolymerization	Device	12	NA	Low	Excellent resistance to common automotive fluids	149
3.	Polypropylene	Crude oil	Commercial polypropylene	Device	8	Squeezing the sponge manually	Low	Highly oleophilic and hydrophobic properties	150
4.	Polypropylene	Light crude	Commercial polypropylene	Fiber/web	10	Compression (simple mechanical action)	Low	Main disadvantage being that they degrade very slowly as compared with vegetable products and are not naturally occurring as mineral products	135
5.	Polypropylene	Light cycle/light gas oil/heavy crude	Commercial polypropylene	Non-woven web	4.5, 4.2, 4.5	NA	Nonwoven PP fabrics face the problems such as high cost for device and long production time	Nonwoven polypropylene sorbents are consolidated fibrous materials, which are different from the conventional textile fabrics. Nonwovens are based on webs of individual fibers. These fibrous webs, which contain small pores, facilitate the transport of liquids into the sorbents and retain the liquids after sorption	151
6.	Gum arabic—polyolefin	Crude oil	Elastomeric gum comprising a block copolymer of ethylene/butylene and styrene and an oleo chemical synthetic wax, the latter being used in amounts of from 1 to 33 weight percent of the mixture	Powder	4 (0.2 to about 1 part per weight part of oil)	Solids can be readily recovered from the water or from a shoreline after they have washed ashore	Low	Physical and chemical stability has enhanced	152,153
7.	PVA/PP	Motor oil	Polyvinyl alcohol and polypropylene under thermoplastic extrusion	Powder	2	Vacuum filtration	Low	The material is highly oriented and biodegradable	154
8.	Swellable porous PDMS	Gasoline oil	Direct curing of a PDMS prepolymer in axylene solution in the presence of commercial sugar particles	Porous polymer	22	Squeeze	High	PDMS oil absorbents show little loss of their absorption capacities and own weights. The absorption process can be finished within tens of seconds	140
9.	PDMS sponge	Chloroform	PDMS and sugar template	Sponge	11	Squeeze	High	PDMS sponge can be elastically deformed into any shape, and it can be compressed repeatedly in air or liquids without collapsing	114
10.	Graphite/isobutylene-isoprene aerogel	Diesel oil	IIR (5 g) was dissolved in benzene (100 ml) at room temperature for 24 h. Graphite (0, 0.5, 1, or 2 w/w%) was dispersed in 10 ml of benzene and then added into the IIR solution. Once graphite was added, stirring was	Highly porous cryogel	22	Squeeze or centrifuge	High	The dried cryogels were sponge like soft materials, with excellent buoyancy and hydrophobicity. In dynamic sorption tests, remarkable swelling was observed. The cryogels were efficient sorbents for the removal of small oil drops dispersed in an emulsion	79

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
			continued for a further 30 min to produce a uniform mixture. Crosslinking agent S_2Cl_2 (0.3 ml, 6 v/w%) was added drop wise to the mixture under rigorous stirring						
11.	Graphene/poly pyrrole foams	Diesel oil	Functionalized graphene oxide (KGO) sheets were covalently assembled with pyrrole and reduced to form a 3D foam structure via a multistep route through the hydrolytic condensation (crosslinking), polymerization reactions, and hydrothermal reduction of graphene oxide (GO)	Foam	108	Wash	High	There is an intrinsic covalent bonding between graphene sheets and the special 3D structure makes way for high sorption capacities of the graphene/poly pyrrole foams	93
12.	Conjugated microporous polymer	Octane	Sonogashira Hagihara cross-coupling of 1,3,5-triethynylbenzene with a number of functionalized dibromobenzenes	Porous polymer	23	Wash	Medium	CMP has open pore structures and excellent surface superhydrophobicity, oils, or non-polar organic solvents can be easily absorbed and separated from water by the CMP without adsorption of water. Absorbency ranging app from 600 to 3300 wt%	155
13.	Polyisoprene–poly (alkyl acrylate) foam	Xylene, toluene, gasoline, diesel and engine oil	The foam was produced via latex vulcanization and cured by steaming using (PMMA), (PBMA), (PBA), sodium dodecyl sulfate (SDS), potassium per sulfate (KPS), and polyisoprene	Foam	2.0–16.6	Squeezing	Low	The modified NR foam can be used as petroleum-based oil sorbent and reused up to 30 times. The foam density and petroleum-based liquid viscosity were directly affected to the oil sorption capacity	99
14.	Polyethylene	Diesel oil	UHMWPE as a flat porous film sheet.	Flat film	315	Vacuum filtration	Low	The experimental results show a good correlation with the pseudo-second order model. The sheets exhibit high oil uptake speeds, requiring less than 2 min in contact with the oil to reach saturation. The sheets fulfill the criteria of high uptake kinetics, high sorption capacity, and high mechanical strength simultaneously	156
15.	Cellulose	Crude oil	Cellulose	Device	18–22	Squeezing	High	In wet environments, the best performing materials were polypropylene and cellulosic fiber. Perlite had comparable absorption efficiencies with polypropylene and cellulosic material. In a dry environment, polypropylene exhibited the best absorption characteristics, whereas all expanded perlites had similar absorbing efficiencies with the cellulosic material for all oils used	157
16.	Cellulose acetate	Crude oil	Cellulose acetate	Device	9	Undergo natural degradation after they are abandoned	High	cellulose acetate utilized to form the hollow core fiber segments (or substitution) has a density of 1.2, somewhat greater than the density of the body of water, the combination of the hollow core fiber segments and spilled crude oil which is drawn into and fills each of these segments	158

(Continues)

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
								by capillary action, must not exceed that of the water density	
17.	Cellulose-*g*-PBMA	Crude oil	Cellulose chloroacetate initiator in DMAc is treated with quantitative DMAP, $FeCl_2$, and BMA [nCell OH:nDMAP:n$FeCl_2$:nBMA =1:1.2:0.70:20 (mole ratio)] at 60°C under stirring, the reaction continued for 4 h under the nitrogen environment and constant temperature to obtain cellulose-*g*-PBMA	Polymeric fiber	96.5	Cell-*g*-PBMA fibers can undergo natural degradation after they are abandoned	High	The oil absorption rate of cell-*g*-PBMA fibers is related to the number of lipophilic groups, oil type, and void network structure of the fibers	159
18.	PMMA-co-PDEAEMA	Hexane, heptane and pet ether	MMA and DEAEMA can be polymerized by free radical polymerization using AIBN at 75°C for 20 h	Nano fibrous membrane	Oil flux of 17,000 L m^2 h 1 (before CO_2 bubbling), water flux of 9554 L m^2 Hour 1 (after CO_2 treatment) for oil/water mixture separation	Membrane filtration	Low	PDEAEMA-based smart membranes that is able to reversibly regulate oil/water wettability of the surfaces. The continuous nanofibrous membranes have an important role in wetting behavior: offering randomly distributed nanofibers to facilitate the roughness of the membrane surfaces, and forming a great number of micropores within nanofibers for further improving the hydrophobicity in air and oleophilicity underwater	160
19.	Nano porous PS	Diesel, silicon, peanut and motor oil	PS fibers by electrospinning method in THF	Film	7.13, 81.40, 112.30, and 131.63 g/g.	Filtration	Low	By controlling the electrospinning condition, polystyrene films with different fibrous diameter and porous morphology can be obtained	161
20.	PVC/PS fiber	Motor oil, peanut oil, diesel oil and ethylene glycol	PVC/PS were taken in 1:1 mixture of DMF and THF and makes a fiber by electrospinning	Fiber	146, 119, 38, and 81 g/g	Filtration	Low	The electro spun PVC/PS sorbent is a better alternative to the widely used PP sorbent for oil spill cleanup due to its high oil sorption capacity, excellent oil/water selectivity, and high buoyancy	162
21.	PVA/PAN composite	Soybean oil	(PVA)/(PAN) Nano fibrous composite membranes were prepared by electrospinning of a very thin PVA nano fibrous layer on the electro spun PAN nano fibrous substrate, followed by remelting PVA nano fibrous layer to form a barrier PVA film by water vapor treatment and chemical crosslinking in glutaraldehyde water/acetone solution	Nano fiber	High water flux with high rejection rate (99.5%) in oil/water emulsion separation	Peristatic pump	Low	PVA/PAN composite membranes possessed high flux (210 L/m^2 h) with high rejection rate (99.5%) in oil/water emulsion separation, and overcame the drawback of easy penetration of the coating solution into the porous substrate in typical fabrication of surface coated composite membranes	163
22.	Chitosan/PAN/nonwoven	Vegetable oil	PAN in DMF is added to alkaline chitosan and cast by electrospinning method to form Nano fibrous material	Nano fiber	High water flux (170 L/m^2 h) 1 at the feed pressure of 130 psi) and good filtration efficiency (>99.9%)	Filtration	High	The membranes can replace the conventional porous membranes and exhibit a much higher flux rate for water filtration, exhibited a flux rate that is an order magnitude higher than commercial NF membranes in 24 h of operation, while maintaining the same rejection efficiency (99.9%) for oily waste-water filtration	164

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
23.	PU-PS core–shell fibers	Motor oil and sunflower oil	PU and PS are taken in 1:3 THF and DMF and form a fiber by electrospinning method	Core shell fiber	64.40 and 47.48 g/g	Filtration	Low	Oil sorption capacities of these fibrous mats were overwhelmingly influenced by the inter-fiber voids among the fibers instead of the intra-fiber porosity. The composite fibrous mats had significantly improved mechanical properties, which overcame the reduction in oil sorbed fiber strength, and resiliency, enabling the composite PU–PS fibrous mats to exhibit an excellent reusability	165
24.	PVDF/Fe_3O_4@PS composite nanofibers	Sunflower, soybean, motor and diesel oil	PVDF solution in DMF/acetone (7/3 ratio) mixed solvent and PS solution in DMF/THF mixed solvent (3/1 ratio) were prepared by overnight stirring at 55°C and at room temperature, respectively. Fe_3O_4 nanoparticles (5 wt w.r.t. the weight of PS) were added in PS solution and the solution was subjected to bath sonication for 2 h to disperse the nanoparticles and electrospinning was carried out to form fiber	Nano fiber composite	35–36.0 g/g	Squeeze	High	The mechanical properties of the composite mats containing both PS and PVDF were found to be significantly improved compared to the neat PS mat. In addition, the incorporation of magnetic Fe_3O_4 NPs in the composite mat helps in the easy recovery of the mats after the oil-in-water sorption process	91
25.	PS nano porous fiber	Motor, bean, sunflower seed oil	20 wt.% PS (Mw = 350,000 g/mol) solution was electro spun from the solvent mixture of THF/DMF with a weight ratio of ¼ and form a Porous fiber by electrospinning	Nano porous fiber	113.87, 111.80, and 96.89 g/g	Filtration	High	The porous PS fibers with smooth surfaces and small fiber diameters have the highest oil sorption capacities, followed by the porous PS fibers with rough surfaces and large fiber diameters	166
26.	Gold nanoparticle–Poly (dimethyl siloxane)	Diesel oil	$KAuCl_4$ + Poly dimethyl siloxane in aqueous media	Foam	125% of its weight	Heating at 300°C	High	The material is ideally suited for removal of organic compounds and oil spills because of its high and fast swelling capacity with longer retention. The fascinating properties of Au nanoparticles could be mingled with those of PDMS to provide a sustainable and practical solution for water treatment. The AuPDMS composite is indeed an attractive material for water-based pollutants and odorous molecules. Act as a sensing material for sulfur impurities usually present in aromatic solvents	167
27.	Graphene oxide/ polyurethane sponge	Oils and organic solvents	Hydrophobic SiO_2/GO nanohybrids using alkyl-rich silicane and construct ternary sorbent by dipping PU sponge into SiO_2/ GO ethanol- water solution	Nano hybrid sponge	80–180	Squeezing	High	The flexible but durable structure of the ternary sorbent exhibited not only excellent sorption ability amounting to 180 times of its own weight, but excellent recyclability which can be fully utilized by only extrusion/sorption	168
28.	Polysilioxane aerogel	Oils and organic solvents	Mixing alkoxysilanes, urea, and the surfactantn-hexadecyltrimethylammonium chloride (CTAC) in a dilute	Gel	80–150	Squeezing	High	The elastic properties of MTMS–DMDMS gels are maintained over a wide temperature range, from LN_2 temperatures up to ~320 °C,	81

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
			aqueous acetic acid solution, and stirring for 60 min at RT to promote hydrolysis					which extends their possible applications to use in harsh environments	
29	CNT/dopamine/ octadecylamine-modified polyurethane (PU) sponge	Hexane, silicon oil	CNT modified PDA and polyurethane were added into dopamine solution and the resulting PU–CNT–PDA sponge was reacted with ODA in ethanol, the resulting sponge was washed with ethanol and dried	Functionalized sponge	29–34.9	Squeezing	High	Mechanical strength of the as-prepared PU– CNT–PDA–ODA sponge is improved due to the structural reinforcement of CNTs anchored on the sponge skeleton. The sponge exhibited outstanding features: water repellency, superhydrophobic stability with temperature and in corrosive solutions, high mechanical strength, excellent elasticity and high selective absorption capacity	169
30	CNT/PDMS-coated PU sponge	Hexane	Coating method for the fabrication of a superhydrophobic and superoleophilic carbon nanotube (CNT)/poly-(dimethylsiloxane) (PDMS)-coated polyurethane (PU) sponge	Functionalized sponge	15	Vacuum filtration	High	CNT/ PDMS-coated PU sponge also allowed the effective separation of surfactant-free water-in-oil emulsions with high separation efficiency (oil purity: >99.97%)	170
31	Fe_3O_4/dopamine-coated PU sponge	Hexane	Fe_3O_4 nanoparticles were dispersed ultrasonically in ethanol and introduced drop-by-drop in the PU sponge and immersed into dopamine solution. After that, the magnetic sponge was modified with an ethanol solution of FAS-17 at RT	Functionalized sponge	10	Mechanical squeezing		The magnetic PU sponge displays excellent superhydrophobicity and superoleophilicity, and more favorably possesses magnetic responsiveness and superior stability against corrosive solutions; it gave outstanding separation performance under magnetic actuation not only for floating oils on the water surface and heavy organic pollutants under water	171
32	Ag/polydopamine coated melamine sponge	Hexane, toluene and octadecene	Dopamine hydrochloride is treated with melamine in aqueous media under O_2 atmosphere. The polydopamine foam is added to $AgNO_3$ solution under ethanol, the composite foam was added into 1H, 1H, 2H, 2H-perfluorodecanethiol ethanol. The superhydrophobic MF foam was finally obtained after thorough drying in air	Functionali-zed sponge	60–95	Mechanical squeezing	Low	Robust superhydrophobic melamine foam with a high absorption ability and good recyclability for oil/water separation	172
33	Crosslinked (eugenyl methacrylate) polymeric microspheres	$CHCl_3$, gasoline, and peanut oil, toluene, diesel and kerosene.	2.0 wt% PVA and 4.0 wt% AIBN and 2 g of eugenol methacrylate were introduced in a 250 ml three-neck flask and were heated to 60°C for 12 h. suspension polymerization	Micro-sphere	31–33	Filtration	Low	Derived from a renewable biomass, eugenol. This is the first polymeric microspheres derived from eugenol. Excellent properties of high oil absorption, fast absorption speed, and reusability	173
34	Poly styrene-co-p-chloromethyl styrene (PSCMS)	DMF, $CHCl_3$, toluene and diesel	Chloromethylstyrene (10 mmol), styrene (30 mmol), and 0.02 wt % of BPO were dissolved in toluene, then the mixture was stirred under N_2 atmosphere at 60°C for 12 h, to obtain the desired copolymer (PSCMS)	Polymeric gel.	82.6, 74.4, 46.7, and 38.1 g/g in DMF, $CHCl_3$, toluene and diesel	Squeezing	Low	The organo-gel attain the maximum swelling capacity during 1.5 h and have high oil sorption capacity make it an attractive sorbent for oil spill cleanup	174

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
35.	Graphene PVDF aerogel	Pump oil, hexane and chloroform	Solvo thermal method	Aerogel	20–70 g/g	Squeezing	High	–	175
36.	Marshmallow like macroporous polysiloxane based gel.	Toluene, mineral oil and chloroform	One pot synthesis method. Mixing alkoxy silanes, urea, and surfactant in acetic acid and hydrolysis followed by gelation	Polymeric gel	Up to 14 g/g	Squeezing	Low	Copolymerization with various types of alkoxysilanes adds functionality to the gels, which allow for the flexible design of surface chemical properties. The introduction of different functional groups allows for changing of adsorption/absorption properties	81
37.	Poly urethane based polymer molecular brush	Crude oil, peanut, toluene, acetone, chloroform, hexane, petroleum ether and methanol	Polyurethane sponge is treated with 1,4 dioxane and stearoyl chloride at 60°C by in-situ chemical reaction.	Polymer molecular brush	17–40 g/g	Squeezing	Low	The grafted PU sponge also possessed high absorption capacity (23 times of the self-weight), high oil retention (93%), high mechanical strength and good recyclability (more than 400 times)	176
38.	Polyurethane sponge	Silicon oil, diesel, gasoline, chloroform, pump oil, hexane and toluene	Dip coating polyamide sponge in to PFA/SiO_2 in an ethanol medium	Sponge	25–65 g/g	Squeezing	Low	The PFA/SiO_2 coated sponge had a faster oil absorption rate, higher oil and lower water absorption than the pristine sponge	177
39.	PET fiber	Hexane	Click reaction between mercapto prophyltriethoxysilane and dodecafluoroheptyl methacrylate. Further in-situ polymerization with PET	Fiber	–	Pumping	Low	Generating durable and robust superhydrophobic surfaces through chemical bonding between the hydrophobic substances and the substrates	178
40.	Polyurethane sponge	Crude oil, lubricating oil, soybean oil, silicon oil, toluene and hexane	Oxidative self-polymerization of dopamine and the reaction of hydrophilic polydopamine (PDA) with hydrophobic octadecylamine (ODA).	Sponge	34.9 g/g	Squeezing	Low	Recovered sponge could be reused to separate oil–water mixture for 150 times, while maintaining its high absorption capacity. The mechanical strength of the as-prepared sponge is improved due to the structural reinforcement of CNTs anchored on the skeleton of sponge	169
41	Polystyrene/fibrous Sponge	Motor oil, silicone oil	PVDF/PS composite package with the sandwich structure was fabricated by alternately one step blow spinning method	Fibrous sponge	359 g/g for motor oil 440 g/g for silicone oil	Not available	High	To enhance the mechanical strength of oil sorbents, the PVDF/PS composite package with the sandwich structure was fabricated by alternately blow spinning. This composite package possesses 2.7 times the tensile strength of the pure PS fibrous sponge	179
42	Lignin based polyurethane foams	Crude oil	Polyurethane foaming process	Foams	6.34 ± 0.27 g/g	Squeezing	Low	–	180
43	PDMS@SiO_2@WS_2	*n*-Hexane, chlorofom	One-step dipping method	Sponge	52.0–112 g/g	Squeezing, continuous vacuum pumping	Medium	PDMS@SiO2@WS2 sponge exhibits separation efficiency over 99.85%. with the merit of non-toxic modifiers and facile method indicating its promising prospect for offshore oil spills cleanup	181
44	CNF aerogels	Olive oil, gasoline, toluene, silicone oil, organic solvents	Super hydrophobic cellulose aerogel was prepared via a CVD method. Freeze-drying, post-cross linking technique for seperation	Gel	77–226 g/g	Squeezing	Low	Modifying of HTMS, Si-CNF/BTCA aerogel showed super-hydrophobicity/super-oleophilicity and adsorption capacity (77–226 g/g) for 30 cycles toward various oil and organic solvents than other oil absorption aerogels	182
45	CNF/PVA montmorillonite aerogel	Pump oil, engine oil, diesel oil, corn oil	Green liquid nitrogen to conduct directional freezing of 1:1:1	Gel	40–68 g/g	NA	Low	Compressive stress of d-MCPMA (0.155 MPa) was dozens of times higher than that of *n*-MCPMA (0.025 MPa) at	183

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TABLE 3 (Continued)

S. No.	Type of oil absorbent	Name of the oil	Preparation	Form	Mass g/g of oil uptake	Desorption method	Cost	Special remarks	Reference
			CNF/PVA/montmorillonite composite aerogel					70% strain. It can carry heavy objects weighing more than 1290 times its weight for up to 2 weeks without changing its shape, which is convenient for transportation and storage	
46	C-dot PDMS, PU sponge	Silicone oil, castor oil	10 mg of PDMS prepolymer solution, 15 mg C-dots were added and stirred for 1 h followed by addition of 1 mg curing agent. The PU sponges were dipped in the C-dot-PDMS solution for 4 h, and the resultant C-dot-PDMS-modified PU sponge was dried at 120°C for 4 h	Sponge	37 g/g for silicone oil, 34 g/g for castor oil	Squeezing manually	Low	This result is seemingly unexpected, because the viscosity of silicone oil (1000 mPa s at 25°C) is greater than castor oil (723 mPa s at 25°C). The more efficient absorption of silicone oil might be due to Si-Si interactions between silicone oil and the PDMS layer	184
47	Silica-aerogels modified with TMCS, MTMS, MEMO	Kerosene oil, diesel oil	Two step sol–gel method by ambient pressure drying and functionalized by mono (TMCS), tri (MTMS, MTES), or organofunctional (MEMO, GLYMO) silanes during surface modification	Porous gel	For S-TMCS Kerosene oil-12.5 g/g Diesel oil-11.6 g/g For S-MTMS Kerosene oil-8.1 g/g Diesel oil-8.7 g/g	Dried at oven	Medium	90% of their total adsorption capacity only within 10 min of exposure. S-TMCS, S-MTMS and S-MEMO exhibited hydrophobic/superhydrophobic nature with contact angle values of 152, 142, and 120° for S-TMCS, S-MTMS and S-MEMO, respectively	185
48	Highly elastic sponge PDMS	Silicone oil, tetrahydrofuran, toluene, soybean oil, tetrachloromethane, petroleum ether	One-pot method	3D porous sponge	559–1955 wt%	Squeezing method	Medium	GO-PDMS sponge is an excellent elastomer, so that its residual adsorption capacity remains at a high level (96.30% of the original capacity) after 30 squeezing recycles. The use of condensing agents releases gas to form a 3D porous structure in the sponge, as a result of cooperation with GO	186
49	Porous silica	Lubricating oil, edible oil, salad oil, diesel oil	Magnetic silica submicroparticles (MSS) were fabricated by using TEOS as silicon source and CTAB as template at room temperature via the template method	Magnetic porous sub-micro particles	11.51, 8.3, 7.14, 5.59 g/g	Vacuum filtration	High	The sample still kept a high water contact angle of 142.1° after absorbing the lubricating oil. Capacities of the modified MPSS for four kinds of oils could be up to 9.42, 7.41, 6.51, and 4.68 times their own weight, respectively, after 20 cycles	187
50	Chitosan, cellulose aerogel modified with TMCS	Diesel oil, organic solvents	Oxidation, crosslinking, freeze drying, cold-plasma modifiation	Gel	7–15.89 g/g for diesel oil 13.77–28.20 g/g for organic solvents	NA	Low	In addition, T-CS-OCA exhibited good mechanical strength and high resilience. During 50 cycles, T-CS-OCA exhibited high cyclic stability. Additionally, the pseudo-second order model was more appropriate for the aerogel's oil absorption process	188

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porous structures with high connectivity and pore sizes above or below 500 μm present the best performance in terms of oil adsorption efficiency and oil saturation time. Li et al.[146] reported the improved oleophilic and hydrophobic properties of PU for oil spill cleanup. The graft copolymerization of oleophilic monomer was performed using (DVB as the crosslinker. The grafting reaction was carried out between PU and lauryl methacrylate (LMA). The optimum monomer, polymerization time, initiator, crosslinker, and monomer weight ratios were found to be 5% and 1.5% for LMA and PU, and the ratio of LMA to PU at 1.5, respectively. The maximum sorption obtained was 46.98 g of diesel and 41.42 g of kerosene per gram of modified PU. The water sorption of the modified PU decreased 50% compared with the blank PU. Liu et al.[147] improved the hydrophobic property of polyurethane foam (PUF) for oil spill cleanup. The PUF with nitrile group was prepared and grafted with octadecylamine (ODA) with a long-chain alkyl group, which is affinitive to oil and/or nonpolar solvents by in situ amidation reaction. The modified foam exhibits higher contact angle (146.3 ± 2.8°) and lower water sorption (0.11 g/g) than the unmodified foam. The sorption capacity of the PUF-ODA for toluene, gasoline, and diesel were 45.3, 35.1, and 33.2, respectively, which have been increased by 20%–40% than the unmodified foam.

10 | LIMITATIONS AND FUTURE PROSPECTS

In recent years, researches are searching for robust new adsorbent materials to provide enhanced oil absorption properties. Considerable work has been performed on nano sorbents, but still their performances are not competitive at the industrial level in spite of its good sorption ability. The sorbent materials for separating the oil/water mixtures are little limited for a large-scale production. The fluoride and other chemical components will create pollutants to the environment and serious threats to our health. Designing an appropriate structure is a good method to acquire desirable wetting behaviors, but affecting the thickness, the pore size, and the mechanical property of the surface. The practical applications of the sorbents are seriously restricted to the fragility and wettability. Therefore, mechanical stability for these materials should be focused with more attention. The extensive application of polymeric nanosorbents with very high and specific sorption capacity has the tendency to adsorb environmental pollutions from contaminated water. To date, polymeric nanosorbents are still dominating the oil adsorbent market due to their ease of processing, high selectivity, and recyclability. Especially metal oxides and carbon-based polymeric sorbents have been shown to have higher adsorption capacities and selectivity. Eventhough, the reported polymeric sorbents in this review may utilize expensive materials, such as nanoparticle and CNTs, its applications in oil spill remediation are still tremendous. However, in the near future extraordinary optimization methodologies may be utilized to explore low cost polymeric nanosorbents at the industrial level. The development of suitable oil loving functional moieties along the polymeric backbone could boost the performance of polymeric nanosorbents. To overcome the cost-effective problems, more attention should be paid on preliminary synthetic steps on large scale with low-cost sources. For further advancement in this area, a high superhydrophobicity, high selectivity, and easy recovery should be ensured for enhanced operations. The valuable properties of nanomaterials in combination with specific polymers molecules will lead to future interdisciplinary research for the new generation of biosensors for water remediation. Along with improvements in existing materials, researchers need to synthesize functional polymeric adsorbents with oil sensing probes in a controlled architecture by means of advanced polymerization methods, which may pave the way toward better adsorption and regeneration.

11 | SUMMARY AND OUTLOOK

As seen from the documented preparation strategies and applications, porous polymeric composites and copolymers can get involved in a complex multidisciplinary science and technology. The recent advances and achievements in the field of polymeric nanocomposites as adsorbents for oil removal are summarized in Table 3.[79,147–187] In order to fine-tune sorption and desorption process, many strategies have been emerged, which in turn most of them are discussed in this review. The modification of the polymeric network by different architecture is also conducive to make versatile and hierarchical polymeric sorbents. The polymeric sorbents exhibit a suitable network structure for oil absorption, where the comparison of oil adsorption by polymeric adsorbents in the references has been summarized in Table 3. Since, polymeric sorbents have a much higher surface area, they have a higher loading capacity compared to traditional sorbents. Polymeric materials and their components exhibit a high oil absorption capability, fast kinetics, and easy recovery from the water surface. In addition, polymeric sorbents have a larger pH stability range, allowing them to be used for the applications that require extreme separation conditions. Adsorption only occurs on the surfaces of the substrate with the diffusion of the adsorbate capacity that is linearly proportional to the specific surface area. However, the weak adsorbate substrate interaction at the surface may cause an easy rebleeding of the adsorbed molecules under a slight external force, which still needs to be further modified and developed in the future.

ACKNOWLEDGMENT

The authors wish to thank for the financial support by the Ministry of Science and Technology (MOST) in Taiwan under the contract number of MOST-108-2221-E-035-054-MY3.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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How to cite this article: Haridharan N, Sundar D, Kurrupasamy L, Anandan S, Liu C-H, Wu JJ. Oil spills adsorption and cleanup by polymeric materials: A review. *Polym Adv Technol*. 2022;33(5):1353-1384. doi:10.1002/pat.5636

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